

Manual

Entry/Information Functions for Shop Floor/Machine Data Collection

AIP-BMD 8.1

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1 AIP Functions - Shop Floor Data/Machine Data

Purpose

The AIP features contained in this function package make it possible to enter data directly in production using shop floor terminals or shop floor PCs.

Implementation considerations

You use the function package if you:

- want to know the current status of each order/operation
- want to know about the progress of the order (time-related and/or quantity-based)
- want to record the activities performed (machine time, labor utilization, quantities) in order to
 - evaluate them, or
 - upload them to a higher-level ERP system.
- would like to monitor machines or would like to enter or track the current status of each machine.
- want to electronically move information such as documents or production notes into Production.

Integration

AIP offers the ability to automatically transfer data from the machine, to record it and to forward it to the server to be posted. Various interfaces are available for this purpose.

Data entered using AIP can be displayed or evaluated in various applications in MOC.

Features

Order-related data entry and posting functions

- Posting functions
 - Log on/log off, interruption and partial confirmation/upload of operations
- Entry of quantities
 - Entry of yield and scrap quantities
 - Entry of rework quantities or open quantities (customizing)
- Recording scrap reasons
 - Entry of scrap reasons

- Reasons for deviation
 - Entry of reasons for deviation in the event of over or underdelivery
- Displaying lists / icons
 - Basic display of workplaces/machines assigned to the terminal or operations logged onto a selected workplace. Presentation in the form of a list or in icon view
- Display of order backlog data
 - Display of selected operation-related information about planned operations or about currently logged on operations: Order backlog data, material components, production resources and tools
- Display of machine data
 - Display of machine master data and shift-related quantities
- Display notes
 - Display of operation-related notes entered on the client
- Modify default values
 - User dialog used to modify the default value for "target quantity"
- Entry of BDE comments
 - Entry of OP-related comments (free texts) added to any particularities documented during the course of the order
- Status log
 - Display of operation or personnel postings made during the current shift

Personnel-related entry and posting functions

- Personnel logon/logoff
 - Personnel logons and logoffs to and from operations or the workplace

Machine monitoring and status recording

- Cycle/operating signal monitoring
 - Automatic monitoring of the production status of machines based on cycle signals or operating signals (requires a connection from the machine to the shop floor terminal).
- Malfunction reason
 - Dialog in which to enter reasons for a disturbance detected by the machine monitoring function
- Modify default values
 - User dialog used to modify the default value for "target cycle" needed for the cycle-based machine monitoring function.
- Manual entry of machine status
 - Manual entry of machine states/statuses

- Status log
 - Display of the machine states/statuses posted during the current shift. Function for the subsequent assignment of reasons to disturbances/malfunctions that had not yet been given one.

Additional licenses might be needed in order to use the functions listed above.

2 Operation of AIP

2.1 Special control and display elements within AIP

Tables

Tables are displayed in a uniform way within AIP. This affects the basic display (workplaces, operations, ...) as well as the selection lists of posting dialogs.

Filter	
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE

Page 1 • Page 2

Provided that information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. By clicking/touching the user can directly switch to another page.

An operation may be selected using the mouse, touch screen, keyboard (arrow keys: '↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Please find the following example: When an operation is logged on, those operations may be selected that are included in the sequencing list or that are planned for the corresponding workplace or group. However, when operations are interrupted, only running operations may be selected.



Scrolling page by page (up or down) in the table.



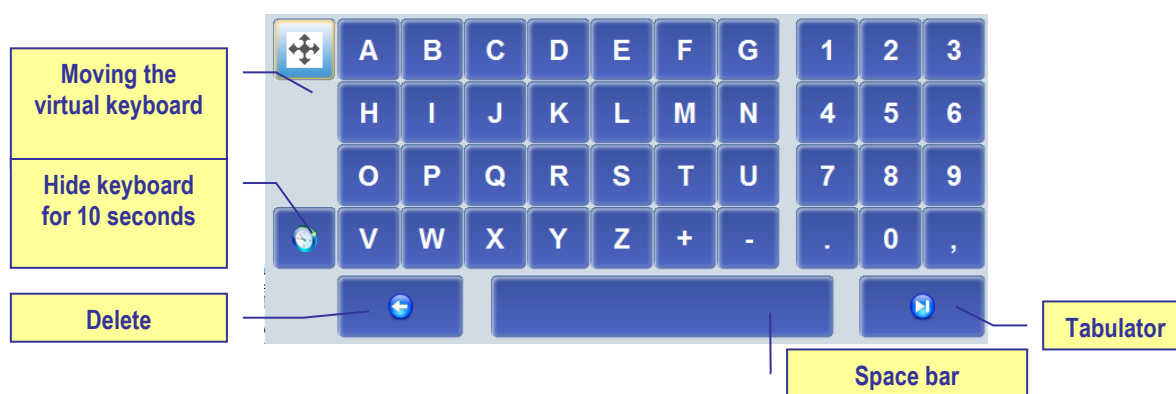
Scrolling to the left or right. Only those buttons are activated that are reasonable for the current situation. This figure shows that scrolling to the left has been deactivated.

Filter

A “table filter” may optionally be displayed (customizing). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is case-insensitive.

Virtual keyboard

The virtual keyboard allows for data to be entered manually via touch screen or a connected mouse. To make it easier for inexperienced users to find the required keys, the numeric key pad is organized like the telephone and letters are aligned in alphabetical order. Consequently, both differ from the computer keyboard which usually is aligned in the “QWERTZ keyboard layout”. The virtual keyboard is displayed automatically as soon as an input field is focused.



The driver needs to be configured respectively for the touch screen to be able to move the keyboard (settings in the control panel of the terminal/PC)! The virtual keyboard only supports the characters "0" - "9", "A" - "Z" and "+", "-", ".", and ",". Other characters or languages are not supported. It is recommendable to use an additional keyboard if texts in other languages have to be entered.

The start position of the virtual keyboard can be defined by a setting in the configuration file keyboard.ini. Subject to the screen resolution, the parameters xpos= and ypos= need to be enabled in the configuration file.

If the virtual keyboard is not to be shown in general, the parameter –t needs to be included in the parameter bar parameters= of the configuration file ctaip.ini.

Date display

AIP supports a country-specific date format in dynamic dialogs. This can be configured in the "control panel", "regional settings", "short date" dialog of the terminal/PC. The following has to be taken into account in this context:

- Years are always four characters long.
- Months and days are always 2 characters long.
- “-“, “/“ and “.” are allowed separators
- Blanks must not be included in the “short date” format, i.e. the <BLANK> separator is not allowed.
- The date separator “.” (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- | | |
|-----------------------|------------|
| • English(USA) | MM/DD/YYYY |
| • Danish | MM-DD-YYYY |
| • Customer-specific 1 | YYYY-MM-DD |
| • Customer-specific 2 | YYYY/MM/DD |
| • Customer-specific 3 | MM/YYYY/DD |

Please note

If the date format does not correspond to conventions a note appears when the program is started and the date format is set to MM/DD/YYYY.

The year is displayed only by two characters in the status bar.

2.2 General description of the posting process with AIP

In general, posting dialogs are divided into several visual views at AIP. These views (partial dialogs) cover the entire screen and only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. This process is described by way of the following example (interrupt operation). The other dialogs can be operated in the same way.

The “interrupt operation” function is executed. This task is started by clicking the “interrupt operation” function from the second toolbar:

Interrupt operation

The “interrupt operation” dialog opens and the first view is displayed. The function that is currently being executed (in this case: interrupt operation) is shown in the header.

The screenshot displays the 'Interrupt operation' dialog for 'Operation 0200 Deburring'. The header shows the title 'Interrupt operation S30002090200 Deburring at workplace 33021'. Below the header, there are three tabs: '1 Enter quantities', '2 Choose status', and '3 Confirmation'. The '1 Enter quantities' tab is active.

The main data area is divided into two sections. The left section, labeled 'General OP data', contains the following information:

- Prod.Order: S30002090200
- Article: SAR-VPO011-M-9005
- Mirror housing right, bla

The right section, labeled 'Posting function that is currently being executed', contains the following information:

- Target qty.: 1200 PCS
- Yield: 660 PCS
- Scrap: 0 PCS
- Completion: 55%

Below the data area, there are two input fields: 'Yield' and 'Scrap'. The 'Yield' field is active and contains the value '22'. The 'Scrap' field contains the value '7'. To the right of these fields, there is a note: 'Quantities that have already been collected (yield, scrap)'.

At the bottom of the dialog, there is a 'Virtual keyboard' with a grid of letters (A-Z), numbers (1-9), and symbols (+, -, ., 0, ,). There are also 'Cancel' and 'Go on' buttons at the bottom.

The first view “enter quantities” provides the user with the possibility to enter the produced yield or scrap quantities. The virtual keyboard is shown or hidden automatically, subject to the active input field.

Quantities can be entered using the virtual keyboard or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). Once all values have been entered in the first view, the next view can be opened by clicking the “next” button.

The “cancel” button is displayed in all partial dialogs and allows for the entire posting dialog to be cancelled/closed at any time.

The next view can be opened either by clicking the “next” button or by clicking another tab (in our example: “select status” or “confirm”). Please note in this context, that no view can be skipped when the views are navigated bottom up (view 1 → view 2 → view 3). This means: if you are in the first view (enter quantities) and you click the third view (confirm), the second view (select status) will be displayed first.

Vice versa, when navigating top down (e.g. from the “confirm” view to the “enter quantities” view), every view may directly be opened by clicking at it. In this case, views are actually skipped. But the “back” button also allows for the views to be opened one after the other (top down).

As long as the dialog has not been confirmed, entered data may be changed at any time by scrolling back and forth.

Interrupt operation 4199LL120010 Assembly at workplace 406457A

1 Enter quantities 2 Choose status 3 Confirmation

Workplace 406457A MOUNT1

Actual status PRODUCTION since 18:40:41

Filter Filter field for the list

Status	Status
1	PRODUCTION
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE
501	COLOUR SAMPLE

Status list

New status 6 START UP Estimated duration 0 Minutes

Comment

Cancel Back Go on

07/12/12 23:14:58

The workplace status that is to be set, once the operation has been interrupted, is determined in the second view “select status”. This status may be chosen from the displayed status list. This list can be restricted using the “filter” field. Once the required values have been entered, the next view can be opened by clicking “next” (in our example it is the last view).

Interrupt operation 4199LL120010 Assembly at workplace 406457A

1 Enter quantities 2 Choose status 3 Confirmation

Operation 0010 Assembly

Prod.Order	4199LL12	Target qty.	30000 PCS
Article	34364-647-345	Yield	6833 PCS
	Seating Plate	Scrap	333
		Completion	22%

Workplace 406457A MOUNT1

New status START UP

Workplace data

Entered quantities

Yield 22
Scrap 7

Quantities posted for the OP.

Staff Number

Input field for the badge number

Cancel

Back

Interrupt OP

07/12/12 23:16:40

The partial dialog “confirm” shows a summary of all values entered so far in the dialog. Provided that the user agrees with the entered data, the “interrupt operation” dialog can be confirmed, once the badge number has been entered. Then the dialog including the data is sent to the server and posted.

If an input field of the dialog is not filled out properly (e.g. a mandatory field is empty) the field is highlighted in red in the corresponding view and focused to enable the user to directly correct the field content.



If a workflow dialog is opened it may directly be exited by clicking the ESC button. This is also the case, if the virtual keyboard is opened. Thus, the ESC button cannot be used to close the virtual keyboard.

3 Basic AIP Display

	→	
--	---	--

In general, *AIP* has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons within the touch screen or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these selection lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

3.1 Basic displays – header and footer

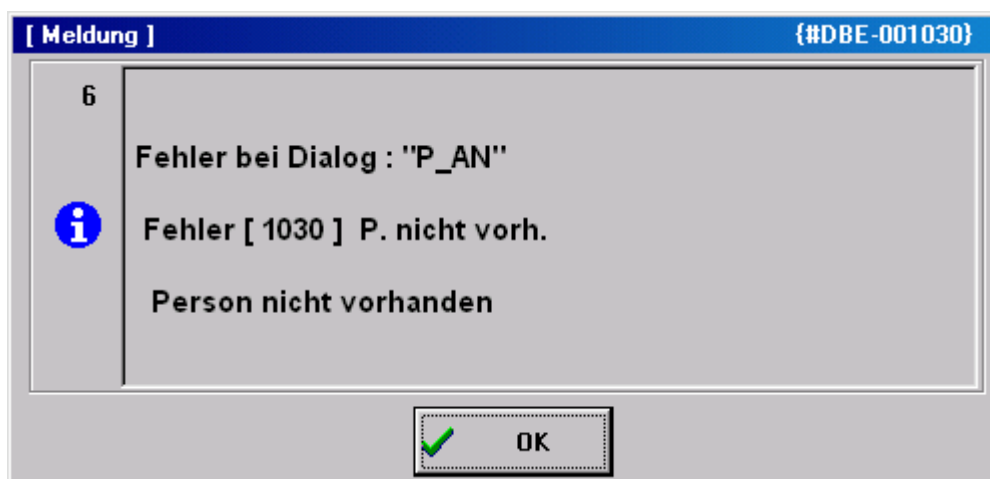
Header



The *AIP* logo is displayed top left of the screen, which may be exchanged by a customer logo after corresponding configuration.

Possible messages (e.g. if a dialog is opened for more than five minutes) are displayed to the right of it.

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

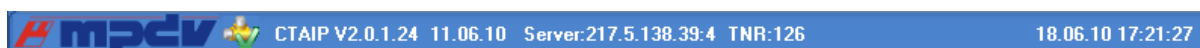


Basic displays

A maximum of 16 workplaces or machines can be assigned to the *AIP* terminal. The individual workplaces can be found within the list area in the order in which they have been assigned to the terminal in MOC.

As regards the basic display of the *AIP* terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured within the terminal configuration at the client. The individual basic views are described in the sections that follow.

Footer

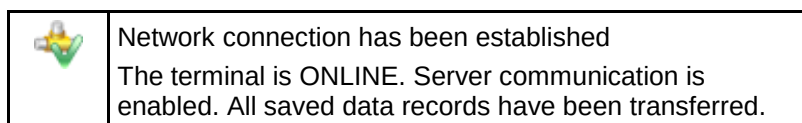


The MPDV logo can be found at the bottom left of the screen. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

AIP status



	Commands are being transferred to the server
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

3.2 Basic display “tabular view“

1st list
Workplaces assigned to the terminal

2nd list
List of registered operations

3rd list (optional)
e.g. list of registered staff

The screenshot shows the AIP (Automated Information Processing) interface. It features three main tables:

- Machines/Workplaces:** A table with columns: Machine/Workplace, Status, Status since, and Note. It lists four entries: 33021 Manual Workplace Deburring, C (PRODUCTION), 40312 Manual Workplace Deburring, C (OUT OF MATERIAL), 406457A Mounting Place 1 (PRODUCTION), and 406457B Mounting Place 2 (PRODUCTION).
- Operations on workplace:** A table with columns: Article, Order, Operation, Target qty., Yield, Scrap, N, and T. It lists one entry: SAR-VPO011-M-9005, S3000209, 0200 Deburring, 1200 PCS, 660, 0.
- Registered persons:** A table with columns: Person, Name, First name, and Advance logon. It lists one entry: 10002630, White, Susan.

Below each table are various control buttons like 'Lock prod.status', 'Modify status', 'Log on operation', 'Partial confirmation', 'Interrupt operation', 'Log off operation', 'PZE', and 'CAQ'. The interface also includes a status bar at the bottom showing the date and time: 07/12/12 22:13:38.

The tabular basic display consists of two or three tables, subject to the configurations made. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional display).

“Machines/workplaces” table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as the designation are displayed.

Status

The status is displayed in color and as status text. Coloring is as follows:

- green: Production
- yellow: Assigned status
- red: Not assigned

Status since

Point in time since the status is available.

For ADE workplaces¹ the point in time refers to the last manual status change, for MDE workplaces it refers to the time when the last status change was identified (for machine connections), to the point in time when the status was changed manually the last time or to the time of the last shift change.

Please note

It is indicated here if the “block production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.



Please note

By way of “customizing” services it is possible to adapt the layout of display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic display of terminals. The software does not allow it.

Provided that the “compensate manual quantities” option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately, once they have been entered but only once lists have been reloaded.

¹ We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an ADE workplace.

"Operations at workplace" table

The second table shows the operations that are currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number*.

Target quantity

Target quantity that is defined for the operation.

Yield

Yield which has already been produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity which has already been produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note, which should be visible at the terminal, has been recorded for this operation in the graphic planning board at the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a line that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured respectively. Which information is displayed in this list depends, among other things, on the workplace configuration.

The following lists can be displayed:

- Staff logged on to the currently selected workplace (BDE)
- Resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow for switching between these lists.

Please note

The registered staff displayed in the third list correspond to the list of the dialog “F5 registered persons...”. Selecting a person in the third list does *not* affect selection of the operation in the list of OPs running at the workplace and, as a result, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic display

A toolbar, which may be configured by customizing, is assigned to each list in the basic display. This makes the purpose of the function clear to the user. The “partial upload/confirmation” function can be found, for example, below the list of registered operations.

In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. change partitioning) can be opened by clicking the corresponding button.



Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

3.3 "Machine overview" basic display

If the “change view” button is clicked in the basic display, the view changes to the following presentation:

Toolbar of assigned machines

Machine ID	Status
60610	Production
60611	Assigned
60612	Assigned
60614	Production
60623	Assigned

Machine information

Workplace/Machine	60611	Status	TOOL FAULT
Short description	ARB-320S	Start	14:00 10.11.2010
Group	SGM2	Duration [hh:min]	15368:30
Divisor	1	Yield	1018 PCS
Target cycle [sec/cycle]	21	Scrap	
Actual cycle [sec/clock]	0	Cycles	
		Note	

Order information

Operation	S61003280100	Target qty.	95000 PCS
Article	SAL-OSI011-K-9001	Yield	1018 PCS
Article Designation	Mirror housing left, crem	Scrap	23 PCS
Remark 1		Target qty. since logon	--- PCS
Remark 2		Yield since logon	0 PCS
Plan. duration	554:10	Deviation [%]	---
		Completion	1%

This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation altogether has three areas:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines as well as to shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. Provided that several orders/OPs are logged on, it can be switched between them using arrow keys that are then shown additionally.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$\text{Displayed partitioning} = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \dots \right]$$

TLG _M	Partitioning of the machine (TLG in mnrlst)
DIV _M	Pulse factor of the machine (IMPFAKT in mnrlst)
TLG _{AGI}	Partitioning of the individual order (TLG in anrlst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview at the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal may get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) at the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time in which the production lock of the machine has not been active). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:

$\text{TargetSinceLogon} = \text{NetRunningTime}[\text{sec}] * \text{Partitioning} / \text{TargetCycle}[\text{sec/stroke}]$

NetRunningTime: Time since logon, in which the production lock has not been set. This calculation does not take into account any breaks that might be defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced "since logon".

Calculation: $\text{Deviation}[\%] = 100\% * (\text{YieldSinceLogon} - \text{TargetSinceLogon}) / \text{TargetSinceLogon}$

Completion

The bar represents the proportion of "yield", which has been produced until now, compared to the "target quantity".

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced by a picture showing a yellow or red oilcan, depending on the maintenance activity that is required:



3.4 "Machines as icons" basic display

This view can be configured as the default display within the terminal configuration at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the "Production" status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon presentation of machines”.

4 BDE and MDE Functions

This document describes the different *AIP* functions used for the Shop Floor Data and Machine Data Collection (BDE and MDE).

4.1 Logging on an operation

When you log on an operation, the following methods are available:

Separate logon of orders and staff	Operations and staff have to be logged on/off separately. You have to enter the staff badge number with the function Log on operation . This is a validation check only. The person must be logged on separately using the function Log on person .
Log person on with order	The OP and the person are logged on in one posting process. You only use the function Log on person if further persons are logged on to this operation.

Make this setting in the terminal configuration on the client (*Log user on with order* option).

Posting process

Select a workplace before you log on an operation. When you then call the dialog, the workplace field is already populated.

Calling the function *Log on operation*

Click the button *Log operation on*.

When the function is called, the user is guided through the input dialog.

Choose operation

Manual entry via keyboard

or

Selection via sequencing list (see the "Notes on the sequencing list")

or

Scan of bar code



With manual entry or bar code scan, the operation is not automatically searched and positioned in the sequencing list.

Choose status

Enter or select the status number that must be set when the operation is logged on.

Staff badge number

The input field is also displayed if you have selected the “separated” option for the "logon" in the terminal configuration (client). The person is not logged on with the order but is only used for validation. For this setting, go to the terminal configuration: *authorizations on the terminal > log OP on*.

Confirmation Log on operation

You use the button **Log on operation** to log on a new operation. With the OP logon, the run times are booked to the different time accounts and the quantities are posted for the OP.

When the operation is logged on, the *Workplace* field in the backlog of orders is filled with the workplace where the operation is logged on. If the operation has been planned for another workplace, the planned workplace is then overwritten. As a result, the OP is implicitly re-planned. This re-planning does not involve any further actions (e.g. update of the template, rescheduling).

Notes on the sequencing list

You can limit the number of entries in the sequencing list for the specific workplace. The list should not get too long because a long list has a negative effect on response time behavior, operability and search time (recommendation: not more than 50 entries, less with remote connections).

The same is valid for the functions *Interrupt operation*, *Finish operation* and *Partial confirmation*. Here, running operations are displayed.

Filter						
Article	Order	Operation	Target qty.	Yield	N	T
SAL-VGO011-M-9001	S3000210	0400 Cleaning	550 PCS	0 PCS		
SAR-VGO011-M-9001	S3000211	0400 Cleaning	600 PCS	0 PCS		
SAR-VTO021-K-5013	S3000515	0200 Deburring	1800 PCS	0 PCS		
SAL-FMO021-K-7036	S3000506	0200 Deburring	2200 PCS	0 PCS		
SAR-FMO021-K-7036	S3000507	0200 Deburring	2300 PCS	0 PCS		
SAR-VTO021-K-3000	S3000519	0200 Deburring	1900 PCS	0 PCS		
SAL-FMO021-K-3000	S3000508	0200 Deburring	1800 PCS	0 PCS		
SAR-FMO021-K-9001	S3000503	0200 Deburring	1400 PCS	0 PCS		
SAR-VSH021-K-5013	S3000523	0200 Deburring	1600 PCS	0 PCS		

If an operation is already logged on to the workplace, then the system enters this workplace in the input field *Operation*. If you do not want this and if the system should preset the first operation of the sequencing list in this field, then you must make the following configuration:

- Call the application [Dynamic dialog configuration - Fields](#).
- Request data for the dialog "WF_AGL". If it is not the logon dialog A_AN or A_P_AN, then identify the dialog.
- Select the field "ANR".
- Click the *Edit* button (note: this is only possible with terminal-specific or terminal group-specific dialogs. It is not possible with AIPDEF 0 dialogs).
- Enter the value "SETVALUE" (without quotation marks) in one of the field attributes 1 to 8.
- Leave the field *Default* empty.
- Save your changes.
- Change to the application [Dynamic dialog configuration](#).
- Restart the dialogs and then restart the terminals.

Displaying notes and texts

Via configuration in the application [Order types](#), you can use the option *Show OP info when logging on OP* to specify for a specific order type that after a successful order logon on the AIP either

- the notes on the operation with an active option *Visible on the terminal*

or

- the long texts of operations

are automatically displayed. The "OP info" dialog is then opened with the respective active page.

Note: To show the information, the AIP must be connected online to the server.

4.2 Logging an operation off

You use the button *Log off operation* on operation level to log off an operation. The posting of run times and quantities is then finished for the OP. Once logged off, you cannot log on the OP again.

Posting process

Select the workplace and the operation that must be logged off in the main view. When the dialog is called, these fields are then preassigned and cannot be changed.

Calling the function *Log operation off*

Click the *Log operation off* button.

When the function is called, the user is guided through the input dialog.

Yield

Enter the yield quantity that must be posted.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). Enter the produced scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Deviation reason

You can activate an overdelivery or underdelivery check for operations or persons. When the operation is logged off, the system then checks if the quantity recorded has exceeded the overdelivery limit or falls short of the underdelivery limit. If the check has a positive result, the quantities posted are rejected with the error message "Overdelivery" or "Underdelivery".

If only a warning is activated for the check, the user can force the system to accept the overdelivery or underdelivery by entering a deviation reason.

Status

Enter or select the number of the workplace/machine status that is set after the operation is logged off.

Estimated duration

Optionally enter the estimated duration of the selected workplace/machine status in minutes.

Note: Only make an entry here if the status is a downtime status (status <> production).

Comment

Enter an optional comment for the status in this field that is also displayed in the machine history.

Staff badge number

The person entered must be authorized to log off the operation. The setting is made in the HR master data. Go to: Shop floor data > BDE authorizations > Lop OP off.

Confirming *Log off operation*

The operation is logged off, once the dialog has been confirmed. You cannot log the operation on again.

4.3 Interrupting an operation

Click the button *Interrupt operation* to call this function. You use this function to stop collecting times and quantities for an order. The reasons for the interruption can be a quantity upload, a shift change or an interruption of the production for technical reasons.

The process of interrupting an operation and the layout of the input dialog are identical to the ones of the operation logoff. The difference is that you can log on an interrupted operation at any time.

4.4 Uploading a part quantity for an operation

You can use this function to upload a part quantity for the order without interrupting or finishing the running OP. The quantities are booked for the relevant OP and the person that makes the posting. Click the button *Confirm partially* to call this function.

Posting process

Select the workplace and the operation from the list. When you then call the dialog, the system preassigns these fields and the values cannot be changed.

Calling the function *Confirm partially*

Click the button *Confirm partially*.

When the function is called, the user is guided through the input dialog.

Yield

Enter the yield quantity that you want to post.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). You can enter scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Deviation reason

You can activate an overdelivery or underdelivery check for operations or persons. The system then checks if the overdelivery limit is exceeded with the quantity posted. If the check has a positive result, the quantity posted is rejected with the error message "Overdelivery".

If only a warning is activated for the check, the user can force the system to accept the overdelivery by entering a deviation reason.

Staff badge number

The person entered must be authorized to upload part quantities for the operation.

4.5 Log person on

Persons are logged on to or off from a workplace. The logging of persons is therefore made on the workplace level.

You can only log on a person, if an operation has been logged on to the workplace.



You can only use this function to log on persons with single workplaces or production orders.

With group workplaces or overhead cost operations, the persons are logged on via the combined logon of order and persons when the operation is logged on. It must be guaranteed here that a staff badge number can be entered during the posting process.

Posting process

Select the workplace where you want to log the person on or off and click the button *Staff logging*. The system displays a list of the *Registered persons* that are already logged on to this workplace. Close this view by clicking *Close information*.

Calling the function **Log person on**

Click the button *Log person on*. A dialog including only one dialog step opens.

Staff badge number

To identify the person, also enter the staff badge number here.

Other notes

Via customization, you can optionally enter an operator position/function or a premium indicator. You specify in the course of the customization how the value is entered: only a field with a selection list or a separate workflow.

You can configure that an advance logon for the next shift is possible within a configurable time before the start of the next shift. Configure this time in the terminal configuration (tab: *MF functions > Waiting period for advance logon of staff*). But if the last operation is logged off before the end of the current shift, all advance logons are deleted. And you cannot log off a person that is logged on in advance.

4.6 Log person off

Persons are logged on to or off from a workplace. The logging of persons is therefore made on the workplace level. You can only log off a person that has been logged on to the workplace.



You can only use the function to log off persons with single workplaces or production orders. With group workplaces or overhead cost operations, the persons are logged off via the combined logoff of order and persons when the operation is interrupted or logged off.

Posting process

Select the workplace where you want to log the person on or off and click the button *Staff logging*. The system displays a list of the *Registered persons* that are already logged on to this workplace. Close this view by clicking *Close information*.

Calling the function *Log person off*

Click the button *Log person off*. A dialog including only one dialog step opens.

Yield

Enter the yield quantity that you want to post.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). You can enter scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Staff badge number

To identify the person, also enter the staff badge number here.

4.7 Log off everyone

You can use this function to log off all persons in one posting that are logged on to a machine.

You assign the authorization to use this function in the HR master data. Activate the following option in tab *Shop floor data > BDE authorizations > Log all staff off*.

Posting process

The posting process is in general identical to the dialog *Log person off*. The only difference is that you cannot record quantities when using the dialog *Log off everyone*.

4.8 Change workplace/machine status

You can use this function to assign a new status to a workplace/machine. You might need this function during setup of the workplace or in case of a malfunction, for example.

A status should be entered, when the AIP automatically identifies a malfunction and the status changes to “not assigned” (also see “Monitoring of operating signals and cycle time”).

You can configure that only persons can perform a status change that are already logged on. Enable the following option in the HR master: *tab Shop floor data > BDE authorizations > Change only if person is logged on*.

Posting process

Select the workplace where you want to change the status and click the button *Change status*.

Status

Enter or select the status that you want to set for the workplace.



With manual entry or bar code scan, the status is not automatically searched and positioned in the selection list.

Estimated duration

You can enter an estimated time in minutes that the new status will probably take. This field should only be filled for downtime statuses.

Comment

You can enter a comment for the status. This comment can be displayed in the machine history.



If the status lasts on and shifts change, then the comment is only assigned to the MDE log record of the shift before shift change.

Staff badge number

Enter the badge number of the person that changes the status. The number is required for the validation check. The person must be authorized to change the status.

Confirmation of the dialog

Once the dialog has been posted successfully, the new status is activated for the machine.

Other notes

It is possible to change from one status to another, unless the automatic status monitoring has been activated for the workplace/machine (*workplace/resource configuration > tab: MDE configuration > automatic monitoring: cyclic or operating signal*) and the workplace is in status “production”.

If you do not want to show a machine status in the status list (any more), then you must disable the option *Status manually at terminal* in the machine status assignment.

Hierarchical statuses

You can build a status hierarchy as of HYDRA-MDE 7.2. In the status assignment, you store the status number of the direct superordinate status for a lower-level status. All statuses that you cannot assign on the terminal and that are only used to show the hierarchy are called "hierarchy level".

Hierarchy levels are displayed in blue font in the status list.

If you select a hierarchy level and double-click/touch the  button, the list of "lower-level" statuses opens. To get to the next higher hierarchy level, click/touch .

You can only select a status in the status list that can be assigned manually. The following error message is shown, if a machine status change on a hierarchy level cannot be assigned.

4.9 Lock/unlock production status



The button *Lock production status* or *Unlock production status* is only active for a workplace if the current status is not "production" and if one of the following conditions is fulfilled:

- The PCC receiving the machine data of the workplace runs in embedded mode and the workplace is assigned to a terminal configured with "MDE operation".
- The PCC receiving the machine data of the workplace runs in stand-alone mode and the INI configuration [MDE-NOTIFICATION](#) is activated for the workplace.

If the function *Lock production status* is active, the terminal cannot automatically change to status "production" when the terminal identifies signals (pulses or operating signal).

If the production status is locked, this means:

- With an active lock, the automatic change to status "production" is not possible and the status currently set is kept despite machine pulses (e.g. "setup").
- If cycle signals are recorded during the lock via counter inputs configured as "yield", then the relevant quantities are booked according to the [Configuration](#) of *Posting during prod. lock*:
 - as yield,
 - as scrap
 - or not at all.
- Click the button *Unlock production status* to remove the lock.

The production status can also be locked/unlocked when a specified workplace/machine status is set. You can configure the behavior in the configuration [Status assignment](#) for each status that is not "production" (go to: *Master data > Workplaces/machines > Status assignment*).

Logging of the production lock

The manual lock or unlock of the production status is logged as event on the server and can be shown in the [Machine history](#). Note: the logging of the event is only performed if the status is manually locked or unlocked.

But if the production status is locked or unlocked via status change, then this is not documented explicitly as an event and is therefore not shown in the machine history.

If the production status is locked for a machine and the terminal software is restarted, then the production lock is automatically removed after this restart. The changed production lock is not logged.

Authorization check when setting the production lock manually

You can configure that only if a relevant authorization is available, the person is allowed to manually (explicitly) set or remove the production lock.

To do so, activate the dynamic dialog M_PSPERRE. If this dialog is activated, you must enter the staff badge number.

If the dialog is activated, the dialog is displayed when the operator clicks the button *Lock production status* or *Unlock production status*. Once the badge number has been entered, the system checks whether this person is authorized to lock/unlock the production status. The system checks, if the option *Change of production lock* in the HR master data (tab *Shop floor data*) is enabled.

If the terminal is OFFLINE, the terminal configuration specifies the behavior (option *Checking required*).

4.10 Change target cycle

The target cycle is the default value that is checked in case of a machine monitoring based on cycle time. You can use this dialog to change the target cycle for the machine.

Posting process

Select the workplace where you want to change the target cycle. Click the button *Change target cycle*.

New target cycle

Specify the new target cycle in seconds/cycle

Staff badge number

The entry of the staff badge number is optional. If configured accordingly, a validation check is performed for the number. (*HR master data > tab: Shop floor data > Change of cycle/partitioning*).

Confirmation of the dialog

Once the dialog has been posted successfully, the new target cycle is activated for the machine.

4.11 Change partitioning

The partitioning (also called cavity) specifies the number of parts produced per machine cycle (clock). You can use this dialog to change the partitioning. The automatic collection of quantities is then based on this partitioning.

Posting process

Select the workplace where you want to change the partitioning. Then click the button *Change partitioning*.

Partitioning

Enter the new partitioning.

Staff badge number

You can optionally enter the staff badge number. If configured in the HR master data, a validation check is performed for this number (*tab: Shop floor data > BDE authorizations > Change cycle/partitioning*).

Confirmation of the dialog

Once the dialog has been posted successfully, the new partitioning is used for all running operations at the machine.

4.12 Change target quantity

Use this dialog to change the target quantity based on operations (primary quantity unit).

Posting process

Select the operation where you want to change the target quantity and click the button *Change target quantity*.

New target quantity

Enter the new target quantity that you want to store for the operation.

Staff badge number

You can optionally enter the staff badge number. If configured in the HR master data, a validation check is performed for this number (tab: *Shop floor data* > *BDE authorizations* > *Change target quantity*).

Confirmation of the dialog

Once the dialog has been posted successfully, the new target quantity is stored for the operation.

4.13 Information on operations (OP info)

General information



Use the button on operation level to call the OP info dialog². Select the required operation. A dialog opens. The dialog includes several pages that are organized in tabs. The information is only requested from the database when you call the relevant page.

Order S3000209 Operation 0200 Deburring				
Description Notes SF-Comments Documents Tools, Resources Components Progress Resource performance acc.				
Operation	S30002090200	Target qty.	1200	PCS
Article	SAR-VPO011-M-9005	Yield	660	PCS
Article Designation	Mirror housing right, bla	Scrap	0	PCS
Remark 1		Completion	55%	
Remark 2				
Planned duration	06:40			

FURTHER INFORMATION:

- EVERY 750 PIECE: QC ACCORDING TO FORMULAR 823273
- PACKING / TRANSPORT STD-CONTAINER NO 4

Close OP information

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The dialog includes the following tabs:

² This function is also available at other places, e.g. in the *Log operation on* dialog.

Description

Current information on the current operation is displayed.

- Operation (MES order number)
- Article
- Article designation
- Remark 1
- Remark 2
- Planned duration
- Target quantity (of the operation)
- Yield
- Scrap
- Completion (in %)

At the bottom, the long text assigned to the operation is displayed.

Notes

The tab *Notes* displays the notes entered on the client (usually via the *Graphic planning board*) if the option *Display on terminal* is enabled.

The table shows all notes of the operation that are configured to be displayed on the terminal. The list is sorted by the editing date. The most recent note is on top and is displayed when you call the view.

If you click/touch a note of the list, the complete text is displayed in the field below.

4.13.1 SF comments

The tag *SF comments* displays the comments recorded during the Shop Floor Data Collection (BDE) or you can record new comments.

The comments recorded are displayed in the *Order information* dialog on the client or can be forwarded via escalation management, e.g. by e-mail.



If an SF comment is recorded for an operation that has been merged on the MOC, then the SF comment is only relevant for this merged operation. The SF comment is not transferred to the single operations.

You cannot record SF comments for the single operations because the single operations are not displayed on the AIP.

With split operations, the SF comment is only stored for the split operation where the SF comment has been recorded. The SF comment is not transferred to the split master.

4.13.2 Documents

You can display documents, graphics or other files on the AIP, which are listed in the table in tab *Documents*.

Select an entry in the list and touch/click the button *Open document*. The file is downloaded to the AIP and displayed using an internal viewer or an external application (according to the file extension).

Internal viewer

Supported formats for the internal viewer: txt, ini, avi, tif, tiff, jpg, jpeg, dcx, eps, ico, pcx, pcc, png, ppm, pgm, pbm, tga, vst, afi, wmf, emf, bmp.

Supported formats for external HTML viewer: htm, gif, wmv, mpg,

External applications

You must install external applications if other file formats (file extensions) than the ones mentioned above are used (e.g. PDF files). The customer is responsible for the installation.

http links as document references

You can also pass http links to a browser for display without having to download a file beforehand. Use a path with the "http" schema (paths are configured on the client via *System administration* > *System settings* > *Paths*).

These links are displayed using the internal HTML viewer provided by the *AIP*. The file extension does not affect the selection of the viewer.

Also the default browser configured in Windows can be used for the display. To do so, configure the following option in the "hytnrcfg.ini" file:

[Terminal->USR 0]

HTTPBrowser=standard

This setting is not recommended for an *AIP* with touch screen, because the operation of a browser can lead to problems.

4.13.3 Tools, Resources

The tab *Tools, Resources* displays the production resources and tools required for an operation.

Note: Documents are displayed separately in tab *Documents*.

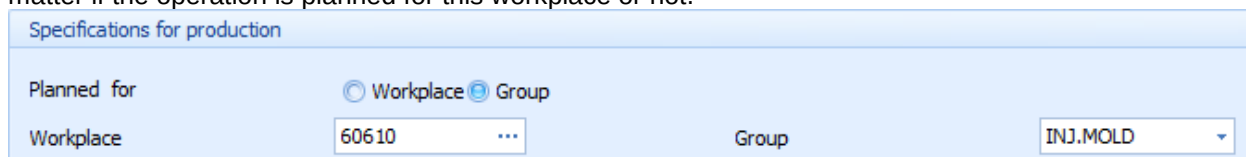
4.13.4 Components

The tab *Components* displays the material components required for the operation.

4.13.5 Progress

The tab *Progress* displays information on the status of the different operations of the order that includes also the currently the selected operation. The below-mentioned data is shown:

- Order
- Operation
- Operation designation
- Color of the status according to the control indicator of the status:
S – (no color), V – gray, L – light green, U – yellow, E – green
- Status (text) of the operation
- Target quantity of the operation
- Quantity unit of the operation
- Yield that has been posted so far for the operation
- Quantity unit of the operation
- Workplace that is assigned to the operation according to the order management. It does not matter if the operation is planned for this workplace or not.



- Group the workplace is assigned to according to the workplace configuration.

4.13.6 Resource performance accounts

In tab *Resource performance accounts*, the following information is displayed for the resource performance accounts (RPA) 1 to 11 of the current operation:

- RPA abbrev.

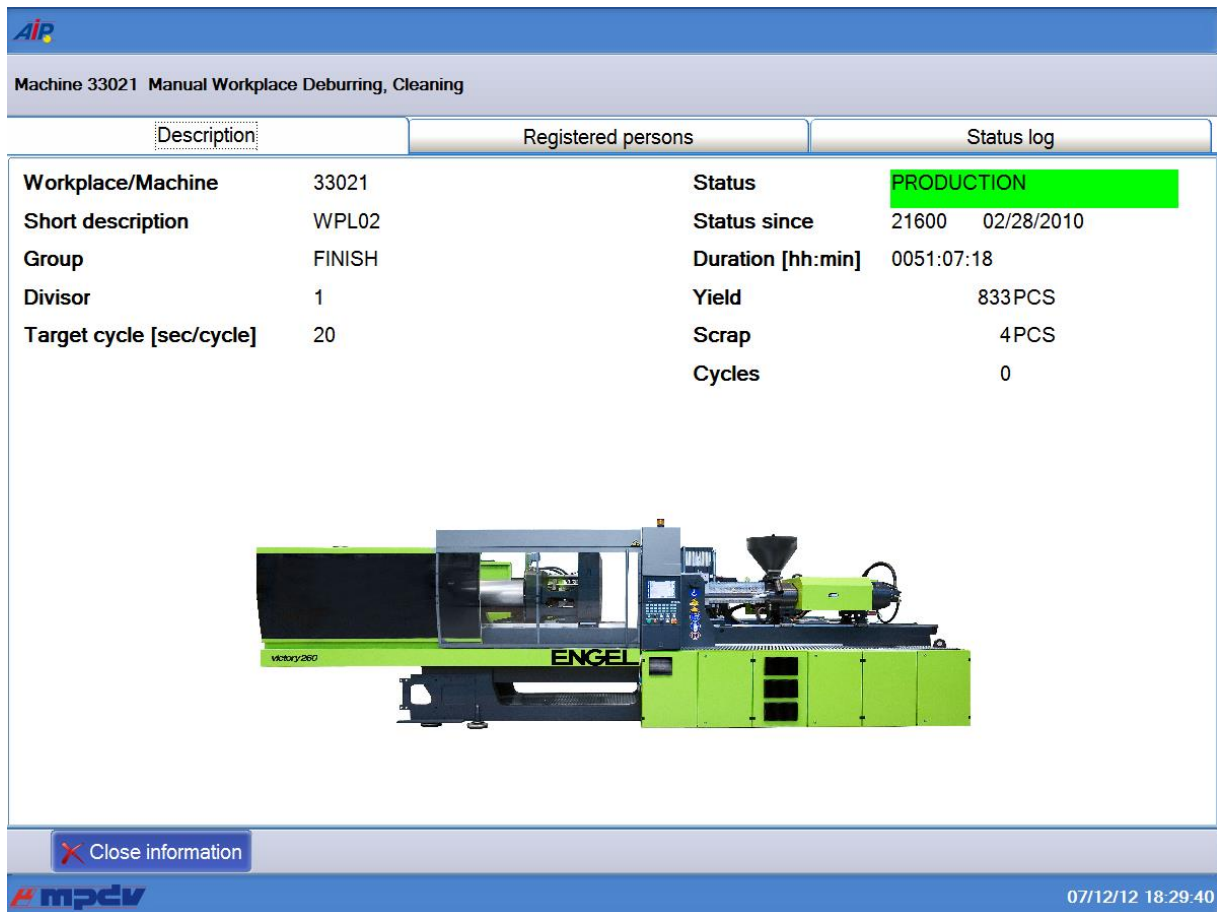
- RPA designation
- Posted duration in hours:minutes
- Total duration = total of all times of RPA 1...11

The durations are displayed in a graphic (to the right).

4.14 Machine information (machine info)

The machine information dialog provides the views and functions listed below.

4.14.1 Description



Machine 33021 Manual Workplace Deburring, Cleaning

Description		Registered persons		Status log	
Workplace/Machine	33021	Status	PRODUCTION		
Short description	WPL02	Status since	21600	02/28/2010	
Group	FINISH	Duration [hh:min]	0051:07:18		
Divisor	1	Yield	833PCS		
Target cycle [sec/cycle]	20	Scrap	4PCS		
		Cycles	0		



[Close information](#)

mpdv 07/12/12 18:29:40

The tab *Description* shows information on the machine/workplace

Workplace/machine

Number of the workplace/machine according to configuration

Short name

Short name of the workplace/machine according to configuration

Group

Group the workplace/machine is assigned to according to configuration.

Partitioning

The current partitioning and the incoming cycles are used to calculate the number of parts per cycle posted for the workplace/machine.

Target cycle

Current target cycle used to monitor the workplace/machine.

Status

Current status

Status since

Point in time that specifies the beginning of the current status.

Duration

The duration specifies how long the current status has been available.

Yield

If the workplace has been configured as MDE workplace, this field shows the yield that has been posted in the current shift up to now.

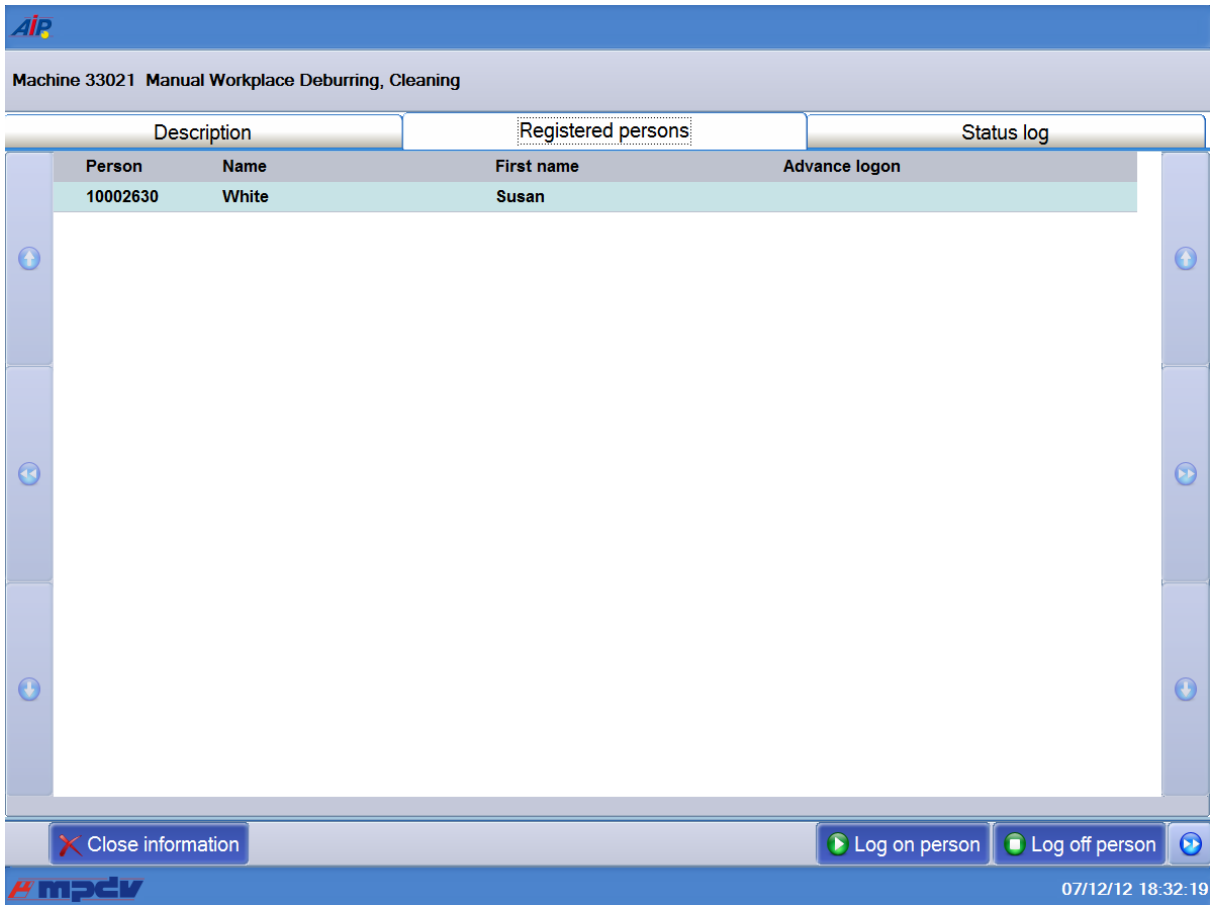
Scrap

If the workplace has been configured as MDE workplace, this field shows the scrap that has been posted in the current shift up to now.

Cycles

If the workplace has been configured as MDE workplace, this field shows the cycles that have been recorded in the current shift up to now.

4.14.2 Registered persons



Description		Registered persons	Status log
Person	Name	First name	Advance logon
10002630	White	Susan	

Close information Log on person Log off person

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This overview shows the staff currently logged on to the workplace.

Use the following buttons to perform postings for staff:

- [Log on person](#),
- [Log off person](#) or
- [Log off everyone](#) (on the next button page)

Click *Close information* to get back to the main view.

4.14.3 Status log

Machine 33021 Manual Workplace Deburring, Cleaning

Description Registered persons Status log

Shift date 03.01.2010 Shift 1

☐ Display statuses without reason only

Event	Date	Time	Duration	Order	OP	Person	Name
Production	16.03.2010	06:00:00	04:10				Reißfelder
OP logged on	16.03.2010	06:00:00	08:00	000001177424	0002	9282	Reißfelder
Person logged on	16.03.2010	06:00:00	08:00	000001177424	0002	9282	Reißfelder
Tool defect	16.03.2010	10:10:00	00:13			1256	Reißfelder
Production	16.03.2010	10:23:20	01:01				STEINBEISSER GEORG
General disturbance	16.03.2010	11:25:00	00:16				STEINBEISSER GEORG
Production	16.03.2010	11:41:40	02:18				STEINBEISSER GEORG
Person logged on	16.03.2010	14:00:00	00:00	000001177424	0002	4232	Reißfelder
Production	16.03.2010	14:00:00	00:18				STEINBEISSER GEORG
OP logged on	16.03.2010	14:00:00	08:00	000001177424	0002	4232	Reißfelder
OP interrupted	16.03.2010	14:00:00	00:00	000001177424	0002	9282	Reißfelder
Person logged off	16.03.2010	14:00:00	00:00	000001177424	0002	9282	Reißfelder
Stamp change	16.03.2010	14:18:24	00:05			9578	STEINBEISSER GEORG
Production	16.03.2010	14:23:57	00:03				STEINBEISSER GEORG

Page 1 • Page 2

Change Status

Close information

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At first, the table shows the following events of the current shift:

- All machine/workplace statuses
- Production lock (set manually, removed manually)
- Postings relating to orders (OP logged on, OP interrupted, OP logged off)
- Postings relating to staff (person logged on, person logged off)

Use the buttons  and  to scroll back and forth shift by shift. You can configure for the specific terminals how many shifts you can scroll back and forth. Use the following option in the hytnrcfg.ini (3 shifts by default):

[MDE->Options 0]

MSTAT.SKNRRANGE=3

In the above example, the screen shows shift 1 if you click  in shift 3. Scrolling back has the same behavior.

Subsequent assignment of reasons

If the machine monitoring function is active, the status switches to the “not assigned” status in case of an absence of signals. When the malfunction has been removed, the machine immediately switches into the production status. The operator cannot enter a reason for this status.

In the status log, you can now list these statuses without reason and subsequently assign a reason to the statuses.



Statuses without reason are statuses that the system has automatically set from “not assigned” to “production” in the course of an active machine monitoring. The production indicator “general disturbance” is assigned to this status and the relevant time is posted for this status.

If an operator manually sets a status, this status usually has a reason assigned, also if the operator uses the production indicator “general disturbance”.

If you enable the checkbox *Display statuses without reason only*, you can filter the list and directly show the statuses without reason only (i.e. automatically assigned reasons). These statuses can then be changed. Note: When you have changed the selection, you must reload the list using the green arrow button!

The *Change status* button gets active when you select a status without reason. You cannot change statuses set manually by the user. The button is grayed out (disabled).

If you click the *Change status* button, you can subsequently assign a reason to this status.

Note

Restriction: Only the MDE data (log records) is changed when you subsequently assign a reason. Postings of other objects (e.g. BDE log records) are not changed.

You cannot subsequently assign reasons to “Short-term disturbances/malfunctions”.

4.15 Merged operations

General information

You use these functions to create or to end “**merged operations**” (MOP) on the AIP terminal. A merged operation includes a group of single operations. The system records the time for the merged OP and proportionally distributes the time to the single OPs. Each person can manage a maximum of one merged operation.

You require a license to use the functions for merged operations on the AIP.

To merge operations, you must click the button *Log MOP on* in the workplace screen. Merged operations are logged off and interrupted like single operations using the *Interrupt operation* or *Log operation off* buttons.



With merged operations created on the terminal, you can only perform logon/logoff/interrupt postings on the terminal. The same also applies to the single operations included in a merged operation. Postings on the MOC are not possible.

Also note that there are some restrictions for merged operations that are described in the documentation [MBL_CollectiveOperationProcessing.pdf](#). Not all postings are possible that are possible with normal operations.

4.15.1 Log merged operation on

You can combine up to 20 operations to form one merged operation. On the AIP, only one operation (the new merged operation) is then displayed in the list of running operations..

Posting process

Select a workplace, before you log on a merged operation. When you then call the dialog, the workplace field is already populated.

Calling the function *Log MOP on*

Click the *Log MOP on* button in the workplace dialog. The user is navigated through the dialog.

Choose status

Enter the status number. Optionally, you can also enter the estimated duration and a comment.

Choose operation

Select all single operations from the list that you want to integrate in the merged operation. Manually select the operations in the list as usual.

When you have selected the single operations, click the button *Add operation*. All selected operations are displayed in green in the list:

Logon merged operations

1 Choose status 2 Choose operation

Filter

Article	Order	OP	OP Designation	Target qty.	Yield
SAL-VGO011-M-9001	S3000210	0400	Cleaning	550	0
SAR-VGO011-M-9001	S3000211	0400	Cleaning	600	0
SAR-VTO021-K-5013	S3000515	0200	Deburring	1800	0
SAL-FMO021-K-7036	S3000506	0200	Deburring	2200	0
SAR-FMO021-K-7036	S3000507	0200	Deburring	2300	0
SAR-VTO021-K-3000	S3000519	0200	Deburring	1900	0
SAL-FMO021-K-3000	S3000508	0200	Deburring	1800	0
SAR-FMO021-K-9001	S3000503	0200	Deburring	1400	0
SAR-VSH021-K-5013	S3000523	0200	Deburring	1600	0

Operation No.

Staff badge number

31.07.14 11:02:19

Staff badge number

Enter the staff badge number here. With merged operations, the person is always logged on with the operation, i.e. a separate staff logon is not possible.

Confirmation via *Start merged operation*

The specified operations are immediately logged on. If the terminal is ONLINE, the system makes a validation check for all entries.

When the merged operation is successfully logged on, an entry in the order overview is generated. This entry includes the characters "**SAM-**" for merged operation and the badge number (e.g. **0160**). The merged operation of the example would therefore be: **SAM-0160**.

4.15.2 Interrupt/log merged operation off

You interrupt or log off a merged operation like single operations. Select an operation named **SAM-
<badge number>** from the list of running operations.

Posting process

Select the workplace and the merged operation that you want to log off.

Calling the function

Click the button *Interrupt operation* or *Log operation off*.

Log merged operation off

Enter the badge number of the person who performs the logoff.

Choose status

Enter the status number. Optionally, you can also enter the estimated duration and a comment.

Choose operation

If the entry of quantities is configured for merged operations in the terminal configuration, a list of the different operations is displayed. To upload part quantities, you can select the relevant single operation, enter the required quantity and click the button *Partial confirmation*. The system does not immediately upload the quantities entered here. The quantities are only uploaded when you confirm the dialog by clicking the button *Interrupt/Log off merged operation*. If the dialog is canceled (ESC key or *Cancel* button), the dialog is closed and no data is uploaded. The partial confirmation is discarded.

Confirmation of the dialog *Log off/Interrupt merged operation*

The merged operation is logged off or interrupted when you confirm the dialog.

If the order is properly logged off, the merged operation is unmerged. After interruption, the single operations are available again in the sequencing list.

The quantities are posted according to the configuration and using the part quantities uploaded.

Further notes

- If you do not enter a quantity with the function *Interrupt merged operation*, then no quantities are posted. With the function *Log merged operation off*, the target quantity is booked as yield quantity.
- If you interrupt a merged operation, the system interrupts all single operations assigned to the merged operation. As a result, the sequencing list does not include the merged operation after interruption, but the single operations. If you want to log on a merged operation again, you must again select or assign the single operations.

5 Specific Features of Machine Data Collection

The functions described in the paragraphs that follow are only active at workplaces meeting the following requirements:

- The machine/workplace has been configured as “individual workplace”
- The machine/workplace is assigned to a terminal, which has been configured in the “MDE” operation mode

These functions are not available at group workplaces.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

5.1 Shift automatic

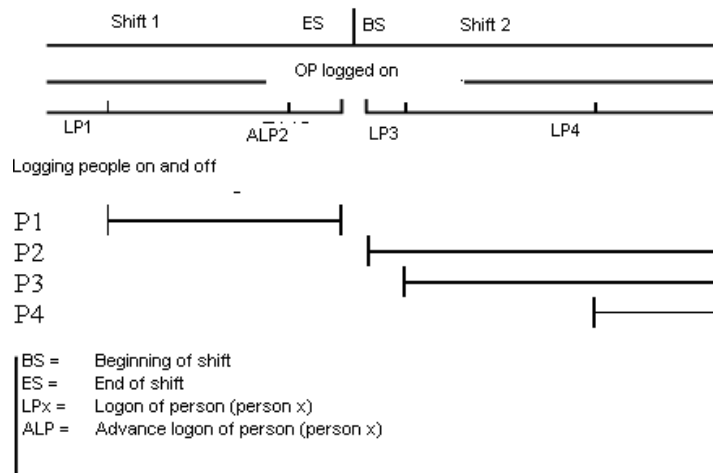
A shift model has to be assigned to each workplace/machine. Due to the information given by this shift calendar, the terminal is able to determine automatically the beginning and end of shifts for its assigned machines.

By the shift automatic option, functions are activated that ease data collection as well as operability:

- the OP that is logged on is automatically interrupted at the end of the shift.
- this OP is logged on at the beginning of the next shift
- Staff can log on in advance to a terminal within a certain period of time prior to the beginning of the shift. When the shift starts the next time, the terminal logs them on to the OP. This period of time is defined within the terminal configuration (option: waiting period for advance logon of staff).

Since data collection at terminals must not be interrupted and it is impossible for all terminals to send all postings simultaneously at the end of the shift, log records are buffered. The buffer is now transferred to the server in short intervals. This process depends on the number of machines defined for a terminal. Postings made during that time are buffered as well.

The following diagram shows a time flow of logons and logoffs during a change of shifts.



Description of the process

- 1) An OP was logged on during shift 1 and its production is also to be continued in the following shift 2.
- 2) Person1 logs on to the workplace.
- 3) Person 2 arrives shortly before the shift ends and logs on to the workplace. Since the logon takes place within the 30 minutes of advance logon time, the terminal recognizes an advance logon.
- 4) The registered OP is automatically interrupted when the shifts end and the staff logged on is logged off.
- 5) The OP, which was interrupted beforehand, is logged on again when the shift starts. Moreover, the persons who logged on in advance are logged on as well.

5.2 Downtime monitoring

The monitoring type is configured in the workplace/machine configuration (MDE configuration tab > Monitoring Type. The following values may be set:

Cyclic monitoring	Cycle time monitoring
Monitoring via operating signal	Operating signal monitoring
No monitoring	No automatic monitoring

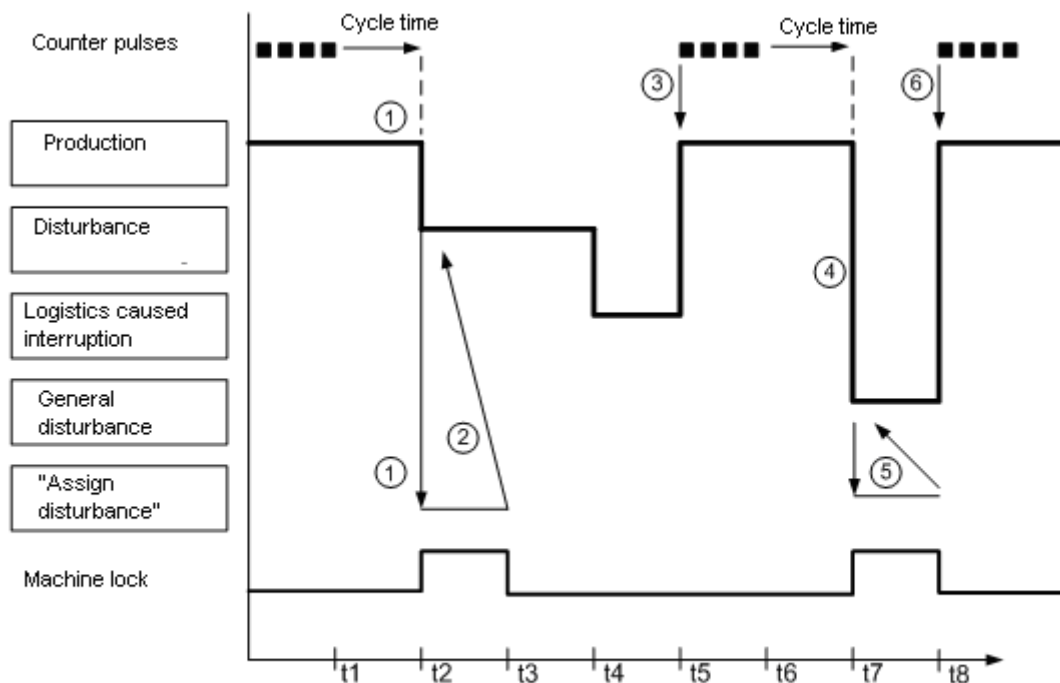
If, after a production phase (cycle time monitoring), the specified target cycle time is exceeded for the logged on order/OP, the terminal detects a downtime and expects a status to be entered. In this case the current terminal status is **“NOT ASSIGNED”**. The same principle applies to operation signal monitoring.

Monitoring of cycle time

When counter pulses arrive, the status is switched to “Production”.

The cycle time is calculated from the target cycle of the OP multiplied by the value entered in the workplace/machine configuration (MDE Configuration tab > Monitoring type > Cycle extension”).

A disturbance/malfunction can only be entered, if the AIP terminal requests it (no production).



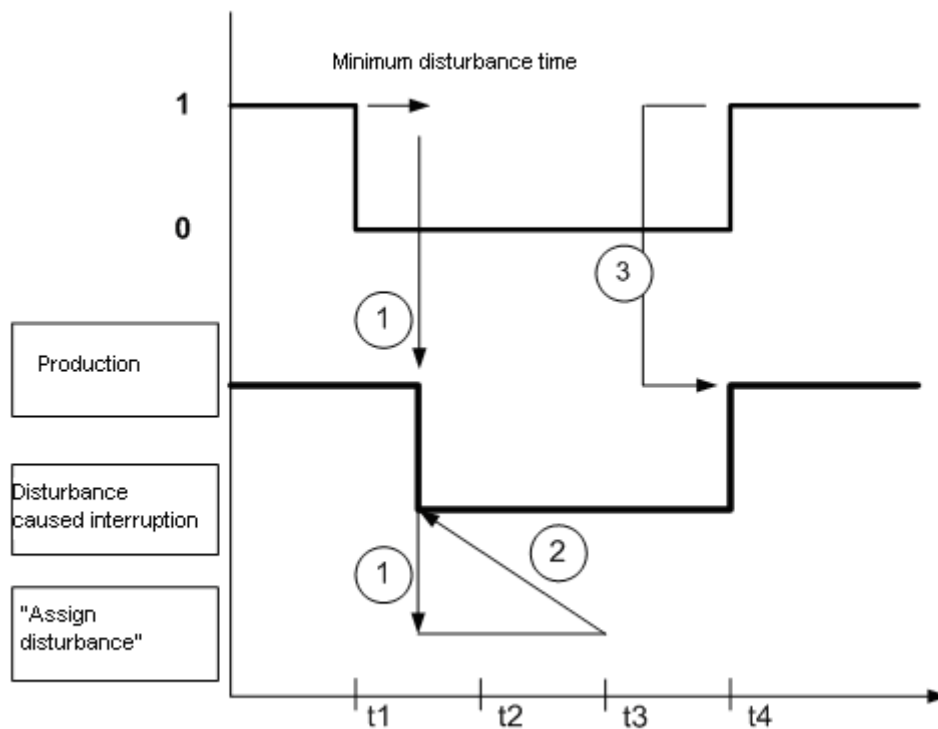
As a part of customizing, it is possible to activate a back posting for this terminal version, in order to record the point in time when the last pulse arrives for cycle time monitoring or the point in time when the operating signal stops for operating signal monitoring.

Operating signal monitoring

By setting the operating signal, status is changed to “Production”.

A disturbance has to last for a certain time, before it will be identified as such and reported. This period is defined by “...MDE Configuration > Monitoring type > Minimum disturbance time”.

A disturbance can only be entered, if the terminal requests it (no production).



As a part of customizing, it is possible to activate a back posting for this terminal version, in order to record the point in time when the last pulse arrives for cycle time monitoring or the point in time when the operating signal stops for operating signal monitoring.

No automatic monitoring

It is possible at all times to define a new machine status. The "Production" status must also be manually assigned.

5.3 Lock production status (production lock)

The "lock production status" button of the "workplaces/machines" section enables the user to prevent the terminal from switching automatically to the "production" status, when new clock pulses arrive from the machine. Thus, switching to "production" is disabled ("production lock").

In case a workplace/machine is unable to switch to the "production" status, the "note" column shows this by "production status locked".

In this case, the current status remains – despite the machine pulses received.

All items produced during the lock are either posted as yield, scrap or not at all, depending on the workplace configuration.

The production lock may be enabled or disabled explicitly by clicking the “lock production status” button.

If the workplace/machine status is configured accordingly (menu: master data > machines/workplaces > status assignment), the production lock may also be set automatically i.e. along with setting a status.

Authorization checking when setting the production lock manually

It is possible to allow the manual (explicit) setting or removing of the production lock only via a respective authorization.

For this purpose, the dynamic dialog M_PSPERRE is to be activated (customizing). The plant ID card (staff badge number) must be entered in this dialog.

If the dialog is active it will be opened, when the worker clicks the “lock production status” button. Once the badge number has been entered, it is checked whether this person is authorized to activate/remove the production lock. This is checked against the “change of production lock” authorization in the HR master (BDE tab).

The OFFLINE performance depends on the terminal configuration (“checking required” option).

Logging of the production lock

The manual activation/deactivation of the production lock is recorded as event at the server and may be evaluated within the machine history.

Please note in this context, that only the event of setting/removing the production lock manually will be recorded.

.

But if the production lock is enabled or disabled implicitly by a status change, no explicit event will be recorded and, as a result it cannot be evaluated at the client.

In case a machine is locked for production and the AIP terminal is restarted, this production lock is automatically disabled after the restart. The process of changing the production lock is lock is neither recorded.

5.4 Machine lock

It can be defined for each status whether or not a “machine lock” is to be enabled as soon as this status is assigned. This can be configured in (status assignment > tab: control > other settings > set machine lock output).

Setting a machine lock leads to an output being set, which may trigger a relay, for example. In this case, the logical output is defined within the workplace/machine configuration (tab: MDE configuration > inputs/outputs > machine lock).

Notes on using the machine lock at DS 100 terminals

The value "1" has to be entered in the "machine lock" field of the configuration of machines/workplaces (tab: MDE configuration > inputs/outputs), provided that the relay is to be set at the DS100 terminal.

The relay is set for all statuses assigned to the "machine lock" as well as for the "not assigned" status.

5.5 Output "target quantity reached"

An output may be set to trigger a lamp, for example, via a relay (to be provided by the customer), when the target quantity of the currently registered operation has been reached.

The logical output is defined in the menu: master data > machines/workplaces > inputs/outputs > target quantity reached.

The target quantity of a machine is checked:

- once the terminal program *AIP* has been restarted
- after changing the target quantity using the corresponding posting dialog
- once an operation has been logged on, off or interrupted
- after posting manual quantities (partial uploads)
- after local quantity events of the MDE module (automatic quantities)
- once the order list has been reloaded

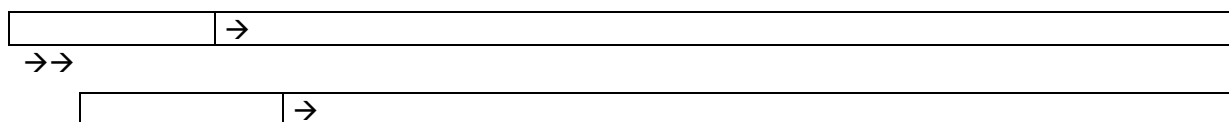
If several operations are logged on to a workplace/machine at the same time the logical output is set as soon as the target quantity has been reached for one of the orders. The signal will be reset if this OP is interrupted.

5.6 Scrap reasons depending on status & production lock

Two scrap reasons can be defined for a status. One scrap reason applying for the active production lock (see section **Error! Reference source not found.** Lock production status (production lock)) as well as one scrap reason applying for an inactive production lock. If a scrap reason is defined for the status that is currently available at the machine the counted scrap will be posted with this specified scrap reason.

Provided that a scrap reason is configured at the counting input, this one takes priority. Consequently, a counter that is assigned to a fixed scrap reason keeps this scrap reason even if another reason is configured for the currently available status.

→



5.7 Setting of outputs subject to the status and posting scenarios

By way of an advanced configuration, digital outputs of the machine may be set if a specific machine status and a defined posting scenario coincide. Further details on this configuration can be found in the document entitled *MDE_digital_outputDepending_on_scenario.pdf*.

Please note the following for the local administration of the machine status at the terminal:

If status changes are posted using PDM or at another terminal the machine status is only transferred when reloading the machine list, provided that this has been configured explicitly by the "FollowExternStatus" option:

HYTNRCFG.INI
<pre>[Tnr Konfiguration 0] FollowExternStatus=on</pre>

When the status is changed by reloading the list, the terminal checks the configured logical outputs and sets or resets them, if required.

6 Barcode Input

A barcode is machine-readable information, which may also directly be generated within the shop floor system. The principal reading devices used are: barcode scanners, swipe card readers or scanners. By default, the terminal supports the barcodes "39", "128" as well as "interleaved 2 of 5 "



Only barcode readers connected to the serial interface (COM interface) can identify barcodes and assign them automatically to the respective input fields at the terminal. This is impossible for barcode readers "connected" via keyboard.

Barcode structure for operations

AAAAAAAAAAAAAGGGGG

Place	Name	Min. length	Max. length *	Example
*	Asterisk	1	1	*
A	Order number	1	12	8
F	Sequence number	0	2	0
G	Operation number	1	4	3
S	Split number	0	2	0
*	Asterisk	1	1	*
Length of BDE order barcode without asterisk (example)				11

* Depends on the configuration of field lengths in HYDRA's basic parameter settings

Barcode structure for machines/workplaces

MMMMMMMMM

Place	Name	Min. length	Max. length	Example
*	Asterisk	1	1	*
M	Workplace/ machine number	8	8	8

Place	Name	Min. length	Max. length	Example
*	Asterisk	1	1	*
Length of machine/workplace barcodes without asterisks				8



To identify a barcode as a machine/workplace number, the barcode must be 8 characters long. If the machine/workplace number is set to “numerical” in the basic parameter settings, it has to be filled up with leading zeroes to reach 8 characters. In case of alphanumeric machine/workplace numbers, they have to be filled up with underscore characters (“_”) to the right.

Barcode structure for machine statuses

NNNNN0

Place	Name	Length
*	Asterisk	1
N	Machine status, with leading zeroes	5
0	Always "0"	1
*	Asterisk	1
Length of status barcodes without asterisks		6

7 AIP configuration of barcodes

7.1 Configuration in ctaip.ini

Readers are configured in section [comports] of the file ctaip.ini:

Section [comports]	
COM1=0 COM2=0 COM3=0 COM4=0 COM5=0 COM6=0	Possible initializations of comports COMx=0 => is not used (by default) COMx=BAR COMx=PSTD PLEASE NOTE! Only starting from reader firmware 69355E COMx=LEGIC COMx=PLG COMx=RFLESER COMx=MSS COMx=SLGB Entry for readers of the type "Schlagbaum" COMx=UKEY U-Key readers Byte 14 and 15 set up the badge/ID card number e.g.: 01010D01 0E020000 04000003 E7175600 03E7 → badge number → 999 1756 → company number → 5974 COMx=KABALEG Kaba Benzing Legic With Bedanet 9580 always COM4 7 bytes are transferred as of byte 15 XXXXXXXXXXXX F:= company number K=badge number COMx=MBB-S6 MBB-S6 reader Attention! Please note! This reader type requires the RS-485 converter to be modified (ECHO=OFF RTS=High) Comports masking has to be set for CLEA + DEUT MBB-S6-MASK=XXXXXXXXXXXX COMx=DRV_UCR New LEGIC Advant PZE/MF reader. required for new PZE readers. The names of these readers include "LGA". Please note: If badges/ID cards are used <u>not</u> complying with MPDV's standard ID, the following parameter must be used for masking in section [Comports-Mask]: DRV_UCR-MASK=.... COMx=PLG-CRYPT New LEGIC Advant ZKS reader required for new ZKS readers. The names of these readers include "LGA". It might be necessary to customize the file plg_crypt.ini for specific customers Set this parameter if Kabalegic and badges/ID cards with MPDV ID are used

	KABALEGM=ON By default, the Kaba Benzing badge ID is used. In this case the parameter is to be commented out As of ctaip #V 2.0.2.27 Read out Kabalegic using configurable search string (e.g. 4F) KABALEG-SEARCHSTRING=XX Correct configuration takes priority (2 places, values 0..9,A..F)
--	--

Section [Comports-Mask]	Example
LESERTYP+'-MASK' =Masking	applicable abbreviations for masking 'T' = Telegram number 'L' = Reader number 'F' = Company number 'K' = Badge/ID card number 'E' = Replacement number 'P' = Check digit (not implemented) otherwise e.g. 'X' = Placeholder (character to be ignored)
Examples: SLGB-MASK=TTTTLLFFFFFFFKKKK BAR-MASK= MBB-S6-MASK=XXXXXXXXXXXXKKKKK	Masking for readers of the type "Schlagbaum": TTTTLLFFFFFFFKKKK = Data string of the reader The badge number may be recorded at every position of the data string Please note!! Impossible for PLG // Status 15 October 2003 DB

The below section has been designed for the configuration of (PLG) Polling Legic Readers or (PSTD) Polling Swipe Card Readers. These readers are mostly used for PZE/ZKS terminals (CT-380) or ZKS terminals (CT-385).

Section [init]	
----------------	--

PlgTimeOut=2	→ TimeOut when starting polling
FreischaltungZyklus=3	→ Cycle for activating/releasing access
MaxComError=15	→ Maximum number of communication errors until access is in "malfunction"
UngueltigZyklus=300	→ Cycle for sending the status "invalid" (invalid badge) in order to re-initialize the reader
InitComError=50	→ Number of communication errors in order to re-initialize Comport
MSSImmediateWrite=false	→ ZKS immediately activates channels for MSS connection (true = not efficient)
Leser-4-Hupe-Aktiv=true	→ Enables an acoustic signal (buzzer/horn) for the status "valid" (valid badge)
LnrErrorPCnt=100	→ Polling is only performed with every n-th attempt if the reader is in the status "malfunction=e". (priority polling)
PinCodeLen=4	→ Length of PIN code input
ComResetTimer=5	→ Minimum cycle for comport re-initialization
RelnitComport=true	Activation of automatic comport re-initialization → by default = false (= without automatic re-initialization)

The below section has been designed to configure MBB readers triggered via a thread in a DLL. These readers are only used for ZKS terminals (CT-385).

Section [MBB-S6-configuration]	
MinMSecDauerOeffner=200 ***** ProcessMessages=true ThreadSleep=20	→ MinMSecDauerOeffner = Minimum duration "door opener" in MSec -> by default [200] *** MBB-S6 - system settings (do not change) ***** → ProcessMessages = Processing of Windows messages while waiting for the application result -> by default [true] → ThreadSleep = Sleep in MSec of MBB-DLL while waiting for application result -> by default [20] range of values [10 .. X] -> changes might lead to increased CPU loads or bad response times

7.2 Configurations in hytnrcfg.ini

Entry	Comment
Section [Terminal->USR 0]	
SuppressBarcodeError=On	Suppresses messages such as 'Barcode ... is wrong', etc. Required if the barcode is processed in the script and normal identification processes cannot identify the barcode
OnBarcode=P_AN	If the AIP basic screen is opened and a barcode is read in, the dialog P_AN opens instead of the dialog M_MST. The scanned badge/ID card number is transferred.
BarcodePrefixChar=\$	Configuration of an alternative separator for barcode prefixes. Can be used if a dot not used as prefix identifier is included at the third place of a real barcode.
ON-KNR-CODE=<Dialog> (By default "M_INFO.PERS")	As of V# 2.0.2.57 Configures the dialog that opens from the main dialog when scanning the staff badge/ID card number using readers such as Legic, Kabaleg, UKey, .

7.2.1 Notes/configurations for concurrent lengths

Entry	Comment
Section [terminal configuration 0] (general configuration) and/or [terminal configuration 2XXX]; (2XXX terminal-specific configuration)	
ANR-COMPLETE-BARCODE-ONLY=TRUE	Option to avoid processing of incomplete order barcodes. (e.g. AUNR,AUNR+AFOLG,...) -> The option is set to <FALSE> by default

RIVAL-BARCODE-LEN-INFO-MSG=FALSE	Option disabling messages indicating that barcodes cannot be processed due to concurrent lengths. -> The option is set to <TRUE> by default
---	--

Please note for configuration: the section may only be inserted provided it does not yet exist.

Sample configuration (for all terminals)

```
[Tnr configuration 0]
ANR-COMPLETE-BARCODE-ONLY=TRUE
BARCODE-LEN-INFO-MSG=TRUE
```

The above configuration prevents incomplete order barcodes from being processed at all terminals (configuration for one terminal with [TNR configuration 2xxx] xxx=terminal number including leading zeroes "0"). If this option is set, only complete order barcodes will be processed. Consequently, valid lengths for order barcodes result from added partial order lengths (Please note: //- ANR (order number) parts with (*) may have the length '0')

```
➔ AUNR + AFOLG (*) + AGNR + UAGNR (*)
➔ AUNR + AFOLG (*) + AGNR + UAGNR (*) + SPLNR (*)
```

The following message is shown if a barcode cannot be processed due to concurrent lengths:

Note [concurrent length]

A barcode (<VAL>)<n>whose field assignment (<IDS>)<n>cannot clearly be defined, has been scanned.<n><n>Processing is only possible<n>if one of the fields indicated is focused.

8 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
-----	-----	---- General ---
00.	00.ABC123	Data not defined, → 00. will be deleted and data "ABC123" will be passed to standard processing
01.	01.OK 01.ESC	Action barcode → Dialog cancelled or ended with OK button or Esc button.
-----	-----	---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Suborder number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16.200	Machine → Acronym <MNR> → Passed to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> → Passed to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> → Passed to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym <EGG:GUT> → Passed to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> → Passed to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
-----	-----	----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---
40.	40.100 40.MONTAGE	Destination → Acronym <ZLO> → Passed to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> → Passed to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→→
43.		Throughput batch number → Acronym <DLL>→→
44.		Alternative batch number → Acronym <CNR:ALT1>→→
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
-----	-----	---- Mainly for the HYDRA-PZE module ---

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> → Passed to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> → Passed to dialog with FGR=1 or FGR=1001
-----	-----	---- Customer-specific barcodes ---
90.		
...		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]

BarcodePrefixChar=\$



If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font "Codedreineun" and prefix "."

Prefix	Barcode	Raw data
10.	*10.123456780100*	ANR = 123456780100
	10._ABCD12340100	ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

8.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR BarKenn91=EGR:GUT	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345" → SAPCNR=12345

Firmly assigned barcode prefixes:

10:ANR
 11:AUNR
 12:AFOLG
 13:AGNR
 14:UAGNR
 15:RMNR
 16:MNR

 17:MST
 18:AUSGRD
 19:AGGGRD
 20:BPOS
 21:LPKZ
 22:SPLNR
 40:ZLO
 41:TPE
 42:CNR
 43:DLL
 44:CNR:ALT1
 45:CNR:ALT2
 46:CNR:ALT3
 47:CNR:ALT4
 48:CNR:ALT5
 49:CNR:ALT6
 50:KNR
 51:PNR
 52:KST
 53:FGR

9 Local Configuration File ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the c:\ctaip directory.



Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

9.1 Basic configuration

Entry	Comment
Section [system]	
Usr=21	Distinct terminal number
Hypath=d:\hydra\ Hypath=/usr/hydra/	HYDRA server path: Windows NT: -> DOS notation, the drive is the local drive of the server on which HYDRA or xMES is installed Unix: -> Unix notation
Hostname=192.9.200.24	Internet address of the server
Offlinetimeout=600	In offline mode, the interval after which online access should be attempted the next time. The interval is specified in seconds
Showcursor=on	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
Loadfile=ctnet\win\ctaip.txt	Configuration file for downloading the application from the server. The path is relative to the server directory (i.e. within "hypath")
Watchdog=on	ON: Watchdog is activated OFF: Watchdog is not activated
Demo=off	'on': Offline demo mode; always off in the production environment
parameters=-t	The -t parameter switches off the virtual keyboard
TMOUT_C=xxx	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_S=xxx	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_R=xxx	Timeout for RECEIVE of the server If not specified, default = 120 seconds

Entry	Comment
TMOUT_F=xxx	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	Configuration of customized barcode prefixes.
BarKenn90=MNR4 ... BarKenn99=ANR3	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.
Section [comports]	
Com1=0 com2=MSS com3=BAR Com3=LEGIC Com4=RFLESER	Assignment of serial interfaces to the connected devices MSS – machine interface BAR, LEGIC, RFLESER – various reading devices
Section [MSS-INIT]	
ZAEHLER= 1 2 3 4 5 6 7 8	Assignment of physical inputs of the MSS (machine interface) to logical counters (ZAEHLER) according to configuration: The first connector (labeled “0” on the MSS) corresponds to the logical counter no. 1 Please note: MSS1 has only 8 inputs If digital inputs are also to be used with MSS1, configuration should be changed as follows: ZAEHLER= 1 2 3 4 IN= 5 6 7 8
IN= 9 10 11 12 13 14 15	Assignment of physical MSS inputs to logical inputs as per configuration: The ninth connector (labeled “8” on the MSS) corresponds to the logical input no. 1
CHARGE= 5 6	For batch recording: Inputs for automatic batch changes In this case, the connectors 5 and 6 are the inputs for the automatic batch change.
MSSZyklusBerechnung=ON MSSZyklusReferenz=0	Activates reading out of cycle time values from the machine interface (MSS). Reference for cycle calculation (smallest time unit of the MSS). The default value is 0, if the parameter is not specified. 0 corresponds to 100 ms. 2 corresponds to 20 ms.
CalculateCycle=ON	The terminal itself calculates the actual cycle. If the connected control does not provide a determined actual cycle, then a calculated actual cycle can be displayed: This calculated value is also available for DS100 terminals. Please note: The calculation cannot provide exact values.
WochenEnde_ProdCheck=ON	This function prevents the weekend automatic from affecting the “production” status and the workplace from being set to status 999. ON is set by default In case WochenEnde_ProdCheck=OFF, the automatism switches to status 999.

Entry	Comment
sFrom999ToNotAttributed=OFF	Only affects HYDRA-MDE machines with operation mode = "no monitoring". If the weekend automatic function is enabled this option prevents the machine from switching to the "not assigned" status when the weekend automatic ends. Reasons: The "not assigned" status may not be set manually for machines that are configured with the "no monitoring" option. The "not assigned" status is normally only set for machines with operation mode = "cyclic monitoring" or "monitoring by operating signal".
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be called to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [PDV]	
Modus=PDV Modus=PDV,BDE	Operation as interface IOP $\leftrightarrow$ DOS terminal (standard) Operation as shop floor terminal with PDV
Terminals=121,122,123	CTDOS terminals "connected" by LAN. Only required with Mode=PDV
PDVTerminalDir=c:\hsrv\spool\ PDVIOPTDir=c:\IOPSim\	Directory for communication with CTDOS terminals Directory for communication with the IOP
For debug purposes only: InfoFenster=100 SlowDown=600	Number of lines of the "current" window (presentation of last PDV actions) Slow down, to make events "visible"
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.
BarcodeWNR=	This field specifies which acronym is entered into the tool number field by the scanner.

Entry	Comment
BarcodeNest=	This field specifies which acronym is entered into the cavity number field by the scanner.
BarcodeNumm=	This field specifies which acronym is entered into the number field by the scanner.

10 Central Configuration File hytnrcfg.ini

This file includes different configurations for all or single terminals at a central place.

Each section is available in a generally accepted version

[section 0].

However, entries included in this section can be overwritten by entries in a terminal-specific section [section <TNR-USER>]

<TNR-USER> = HydraUser = Terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/HYDRA User



The hytnrcfg.ini file is loaded from the server every time the terminal is started.

Section / Entry	Comment
	→→→
[Tnr configuration 0]	
FollowExternStatus=on	Transfer of machine statuses when reloading machine list Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
InstallFonts=on	If this is set to "off" fonts will not be installed during the restart. ON=DEFAULT
OnlyInstallFontsAfterDownload=false	"InstallFonts=on": If true then fonts will only be installed directly after a download. If false then fonts will be installed every time the terminal is restarted. (false = DEFAULT)
InstallTvicport=on	If "off" the LPT driver "tvicport.sys" will not be installed. It is required for HYDRA-ZKS. ON = DEFAULT
[Terminal->USR 0]	

AttachedApplication=First	This configuration checks whether or not an application is connected in Windows that matches the file extension of the document to be displayed from the OP info dialog. If there is such an application, it will be used for displaying the document. If there is no connection, viewers configured in ctaip.ini (→ [ext. software]) and internal viewers will be used. In case, an extension is completely unknown it is attempted to display it as text Different settings may be configured: First → search for connected application first AfterUserViewer → If a UserViewer is configured this one overrides the connected application (also applies for ExcelViewer, WordViewer and PowerpointViewer) Last → Only if no ctaip.ini assignment is found for the file extension, then the connected assignment will be searched for (default). Off → Connected application is never searched.
HTTPBrowser=standard	Viewing of documents (via OP info): If documents are configured with a path of the <u>type</u> "http", the file will not be downloaded to the terminal, but the link will only be transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows will be used.
SupressErrorMessage=70012	Suppress message "material is not planned"
[SignatureRecording->User 0]	
ManualBadgeInput=true	This configuration specifies whether or not the field "user" can be edited in the terminal (by default: no editing) true → activates keyboard input for the "user" field in the terminal

Transparency=255	The signature dialog can also be transparent. 255 → Signature dialog is 0 % transparent (not transparent) 1 → Signature dialog is 99% transparent (maximum transparency) (Default = 155) Available as of CTAIP (V# 2.0.2.25) / CTWIN (V# 7.2.5.99)
ShowPosition=TR	The position of the signature dialog can be adjusted as follows: TL Top – Left TM Top – Middle TR Top – Right ML Middle – Left MM Middle – Middle (Default) MR Middle – Right BL Bottom – Left BM Bottom – Middle BR Bottom – Right Available as of CTAIP (V# 2.0.2.25)
USE_SERVICE_ACCOUNT=1	0 (default) SSO: do not use ServiceAccount (requires the terminal to be started with the "user" domain (SSO)). Please note: ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY	REPORTING_USER_READONLY The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field is read-only. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. REPORTING_USER_CHANGEABLE The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field can be modified. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible.

SIGNATURE_1_LOGON_TYPE=HYDRA	"" / Not set / "EMPTY" There is also an alternative login procedure. HYDRA The tab identifying users via the Windows user is blocked. The HYDRA user must be used for identification purposes. This requires, however, that in the HYDRA HR master all users logging in are created and that the "SSO" option is not set. Otherwise, successful authentication is impossible. ACTIVEDIRECTORY The tab identifying users via the HYDRA user is blocked. The Windows user must be used for identification purposes. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. MIXED_BUT_UNIQUE Either the HYDRA or Windows login procedure is available, subject to whether or not the "SSO" option is set for the registered user in the HYDRA HR master. "SSO" enabled → Windows only "SSO" disabled → HYDRA only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures with the terminal in the area of quality data collection.

10.1 Layout configuration

Entry		Comment
Section and/or	[terminal configuration 0] [terminal configuration 2XXX];	(general configuration) (2XXX terminal-specific configuration)
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE		In case of an error in reading the clock (e.g. after coming out of standby mode), this configuration makes sure that the time is accepted without having to confirm a dialog. Afterwards the terminal time will be synchronized with the server time using a PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON		As of ctaip V# 2.0.2.23 Prevents the warning after restarting the terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default = OFF)

Entry	Comment
NetRuntimeMode=2	As of ctaip V# 2.0.2.50: Alternative calculation of the target quantity since logon: The net run time is not calculated from the times when the production lock is enabled (PSperre=green) but only from the shift times less the shift breaks. Consequently, it can also be displayed, even if the terminal program has been restarted.
Section [QRD-PRINTER->TICKET 0] ;(general configuration) [QRD-PRINTER->TICKET 2xxx] ;(2XXX configuration for a specific terminal)	
COMPLETE-ABSENCE-OF-LOCAL-MNR- DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured <Events>, if it is not available locally => This configuration might be required/necessary for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR- DATA-FOR-EVENT=< Events >	Reloads the order row for the configured <Events>, if it is not available locally ➔ This option has been implemented to access order data within the master data, e.g. when logging an order on.
COMPLETE-...-EVENT=< Events > COMPLETE-...-EVENT=#ALL# COMPLETE-...-EVENT=A_AN A_P_AN	Explanation on the configuration of <Events> ➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event. ➔ <A_AN A_P_AN> restricts reloading of information to the specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.

11 Local Configuration File keyboard.ini

Settings for the virtual keyboard are configured for specific AIP terminals in the keyboard.ini file of the directory c:\ctaip.



Changes to the configuration file will not take effect until after the terminal software has been restarted.

Logic enabling the virtual keyboard:

The AIP terminal shows the keyboard for fields where data can be entered. The keyboard is positioned as described below:

Logic for positioning of the virtual keyboard:

As soon as the activation command has been sent, the keyboard receives the coordinates for the center as well as information on the height and width of the control element to which it is added.

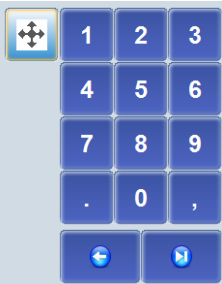
At first, an attempt is made to attach the keyboard directly below the control element. If the lower margin is not sufficient, an attempt is made to place the keyboard directly above the control field. If the space above the control element is not sufficient for the keyboard, an attempt is made to position the keyboard at the bottom margin of the screen. In case the space above the control panel is three times as large as the one below the control element, the keyboard will be positioned at the top margin of the screen.

These are the priorities for horizontal alignment:

- to the right of the control
- to the left of the control
- to the screen margin that is farther from the control

If the "VirtScreenSize" option is enabled, the virtual keyboard is not aligned within the virtual screen but still within the real screen. Consequently, the keyboard may also reach beyond the terminal program.

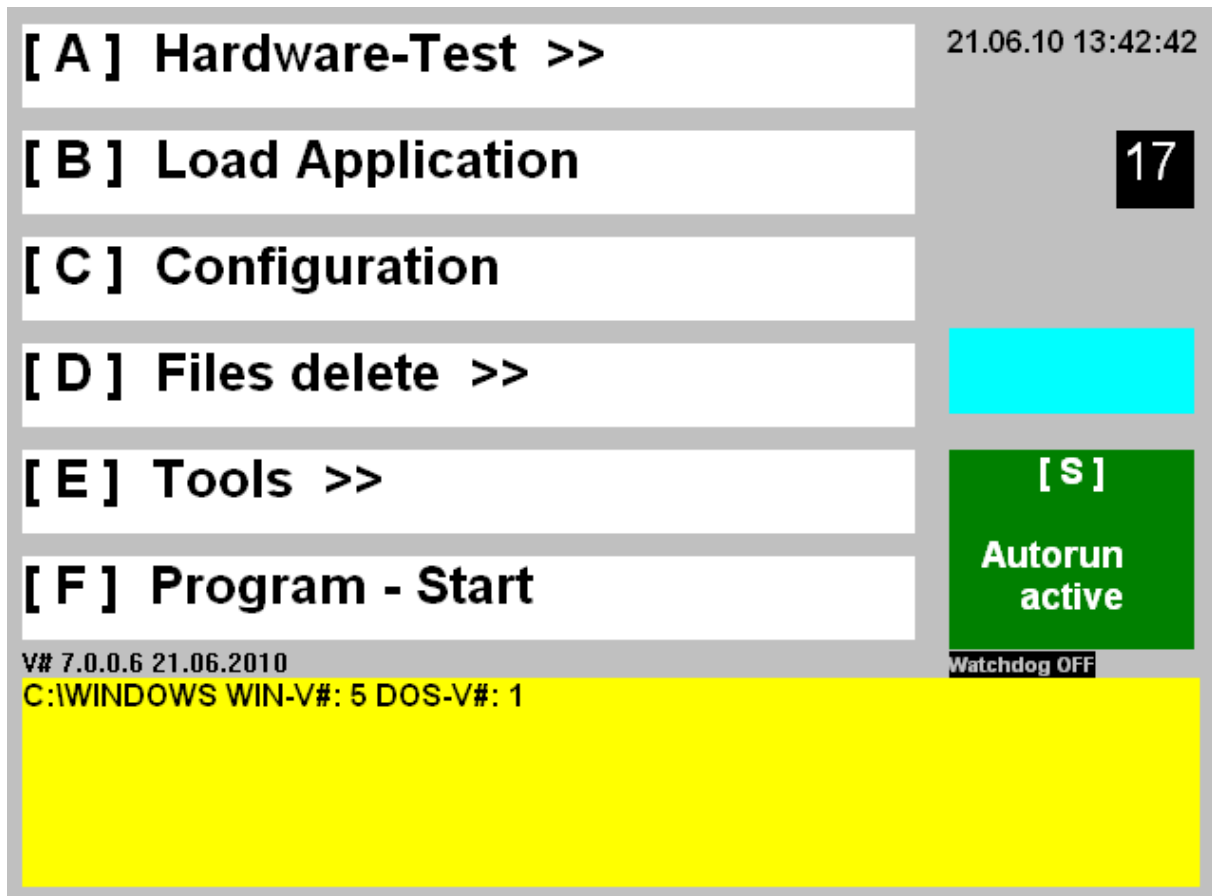
Entry	Comment
Section [User]	General settings

Entry	Comment
Key1=del,8,Delete Key2=<,37,Cursor left Key3=>,-39,Cursor right Key4=<<,36,Cursor at the beginning Key5=/	Definition of additional customer-specific keys displayed in the upper row:  Keys are configured with the following syntax in tasten32.ini within the section [User]: Key<i>=<CH>[,<Code>[,<Comment>]] Key<i>: Key1..Key5 <CH>: Characters displayed on the key and the code of which is sent successfully tested characters: \$=)/?'`{[]}#.,;<>_*~:€äüöÖ²³@ª¿®Çüéääæ/£ø£®½¼©¥ăµ Characters that lead to errors while testing: !"\$%(&,' <Code>: Code that is sent instead of the character code <Comment>: any comment
ButtonShift=ON	Show key "shift" The key has been designed for switching between upper case and lower case letters
ContextSensitive=off	Starting from version 2.0.1.5 of keyboard.exe (30 April 2012), the virtual keyboard automatically shows or hides letters. This depends on whether the current field is numeric or alphanumeric. This button disables the function so that the virtual keyboard is always "opened" ("dropped down"). 

Entry	Comment
HideTime=10 HideTime=0	Starting from version 2.0.1.5 of keyboard.exe (30 April 2012), there is a new button on the virtual keyboard allowing to disable/hide the keyboard for a configurable period of time (by default=10 sec.). Once switched to a new field, the virtual keyboard will immediately appear at the new position even if the hide period has not yet expired. The "hide" button is removed by configuring HideTime=0. 
HideMode=1 HideMode=2 Trace=1	HideMode=0 (by default) means the virtual keyboard appears, once another dialog (e.g. a message indicating "... is being loaded...") is opened and the input focus returns to the dialog. 1: The virtual keyboard is hidden over the indicated period, even though another AIP dialog is opened. The virtual keyboard is shown immediately after switching to another field of the dialog. 2: Like (1) but the virtual keyboard remains hidden even after switching to another field of the dialog. The virtual keyboard is only shown before the HideTime has expired, if the dialog is closed and reopened. Trace=1 enables logging (prot_kbd.txt) for the virtual keyboard. (AIP 2.0.3.25 / keyboard 2.0.2.3)
ScaleMultiplier=0.7	Additional scaling factor reducing or increasing the keyboard. This makes it possible to cover as less information as possible and to keep maximum user friendliness. This setting is useful, in particular, for large screens with low resolution and vice versa. (keyboard.exe V2.0.1.5)
KEYMODE=QWERTZ	Alphanumeric mode based on the German typewriter keyboard (keyboard.exe V2.0.2.2)
Section [SKIN]	General settings
[SKIN] Directory=..\..\..\tastatur\skins Name=mpdv Saturation=0 Hue=0 SkinButton=on	Configuration of the skin (normally, default values are sufficient – the section is not required) Shows a button enabling and disabling the skin

12 Start Menu Inst32

The terminal software is installed on a local hard drive of the terminal or a shop floor PC. After booting the terminal, the following selection menu appears:



The following selection options are available in the selection menu (start by key or clicking)

[A] Hardware Test >>

Submenu for different hardware tests

[B] Load Application

After an update from the HYDRA server, the application programs can be easily updated.

[C] Configuration

[D] Delete Files >>

Submenu for deleting different files (data directories, queues, logs, application)

[E] Tools >>

Submenu for opening or installing different tools

[F] Program – Start

Opening the application program. If nothing is entered, it will be started automatically after 20 seconds.



The <D> and <E> buttons are only used for service purposes.

If no button is pressed, the AIP application starts automatically after 20 seconds. Once the application has started, a few notes are displayed indicating the version number, hardware configuration and other service information. If everything (configuration and terminal) is in order, the display switches to the basic dialogs described in more detail in the sections that follow. If the terminal detects hardware or configuration errors, it alerts the user by sending messages in plain text.

Submenu for opening or installing further [A] Hardware Tests >>

[1] MSS Test

(Hardware test for the machine interface / MSS)

[2] Reader Test

(Hardware test for different reader connections / BAR, LEGIC, PLG, ...)

[3] LAN Test

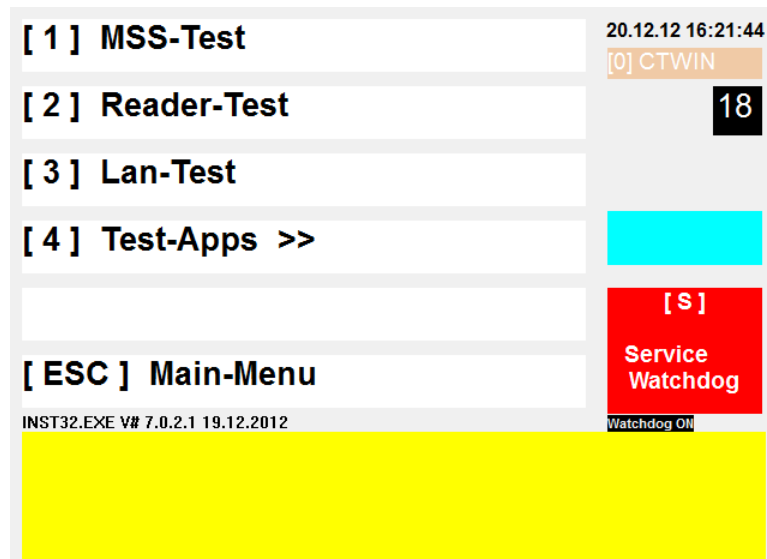
(Hardware test of the network connection)

[4] Test Apps >>

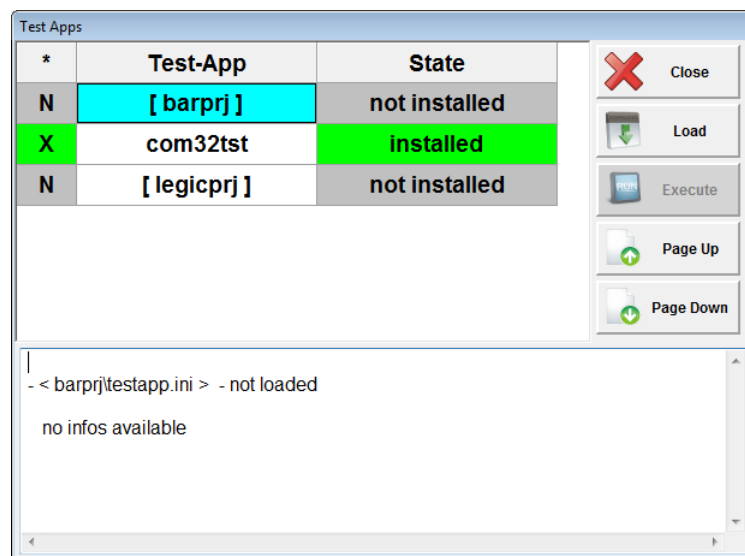
(Test applications of any kind)

[ESC] Main Menu

Back to the main menu



Starting the function “Test Apps“ by the menu item ”[4] Test Apps >>“



When starting the function “test apps“, a directory list of the HYDRA server directory ”./cnet/win/testapps/“ is loaded.

The function creates an entry for each directory in the list that includes a file “testapp.ini”. If this file already exists locally, i.e. it has already been loaded earlier, the entry will be highlighted in green (e.g. **X** **com32tst** **installed**).

The selected “Testapp“ can be reloaded or updated by the function key ”Load“.

The selected “Testapp“ is started by the function key “Execute”.

The list can be browsed by using the function keys “Page Up” and “Page Down”. The function is exited by using the key “Close”.

The configuration file "testapp.ini" is structured as follows.

```
[APP]
name=com32tst
exe=com32tst.exe
param=...          ; optional parameter to transfer call data
[COMMENT]

      MSS Test Program

[/COMMENT] [APP]
```

Submenu deleting or restarting/closing files using [D] Delete files

[1] Delete spool directory

(deleting the ./spool directory of the terminal)

[2] Delete queues

(deleting the queue in the ./spool directory of the terminal)

[3] Delete logs

(deleting the log files in the ./spool directory of the terminal)

[4] Delete application

(deleting the configured application)

[ESC] Main menu

back to the main menu

[1] Spool-Directory delete	24.11.15 15:31:43
[2] Queues delete	[0] CTAIP
[3] Logs delete	Reboot / Shutdown 18
[4] Application delete	
[ESC] Main-Menu	[S] Service Watchdog
INST32USR.EXE V# 7.0.4.3 19.11.2015	Watchdog OFF

The application / Windows can be exited or restarted in the following selection dialog after entering the password "mos6950" and clicking the button .

Function Selection

Submenu for opening or installing further “[E] tools >>“

- [1] **MSS-Loader install**
- [2] **MSS-Loader start**
- [3] **VNC install**
- [4] **HYDRA Fonts install**
(loading and starting the installation package for Font Installation)
- [5] **more Tools >>**
(Backup, Restore, Terminal Upload, Install 3rdParty)
- [ESC] **Main Menu**
Back to main menu

Submenu by starting “[5] more tools”

- [1] **Backup**
- [2] **Restore**
- [3] **Installation 3rd Party**
Please note for installation tool as of version 6.5.2.01:
The "installation 3rd party" function is only possible in connection with hyfserv file server from version 7.2.1.29 onwards
- [ESC] **Main Menu**
Back to the main menu

Menu item “[1] Backup“

The file <hydradir>\ctnet\win\ctaipbackup.txt

or a terminal-specific file <hydradir>\ctnet\win\ctaipbackup2xxx.txt

(xxx is the terminal number) is used by the server for the backup.

At first the system attempts to load a terminal-specific file.

If no terminal-specific file exists, the system will then attempt to load the file ctaipbackup.txt.

This file contains all of the files or registry entries that need to be backed up.

\ctaip*.INI

\ctaip*.cfg

\ctaip\cfg*.*

HKEY_LOCAL_MACHINE\SOFTWARE\MPDV\

A Zip file is created in the terminal and it is stored in the server.

The file is located in the server at:

The backup Zip file is given the name: ctaipbackup2xxx.zip ->xxx = terminal number (terminal-specific for Hydra user 2xxx)

This backup file is then stored in the server under

<hydradir>\custom\backup\ctaip\ctaipbackup2xxx.zip .

Menu item "[2] Restore"

There will first be a query asking whether you would like to run a restore.

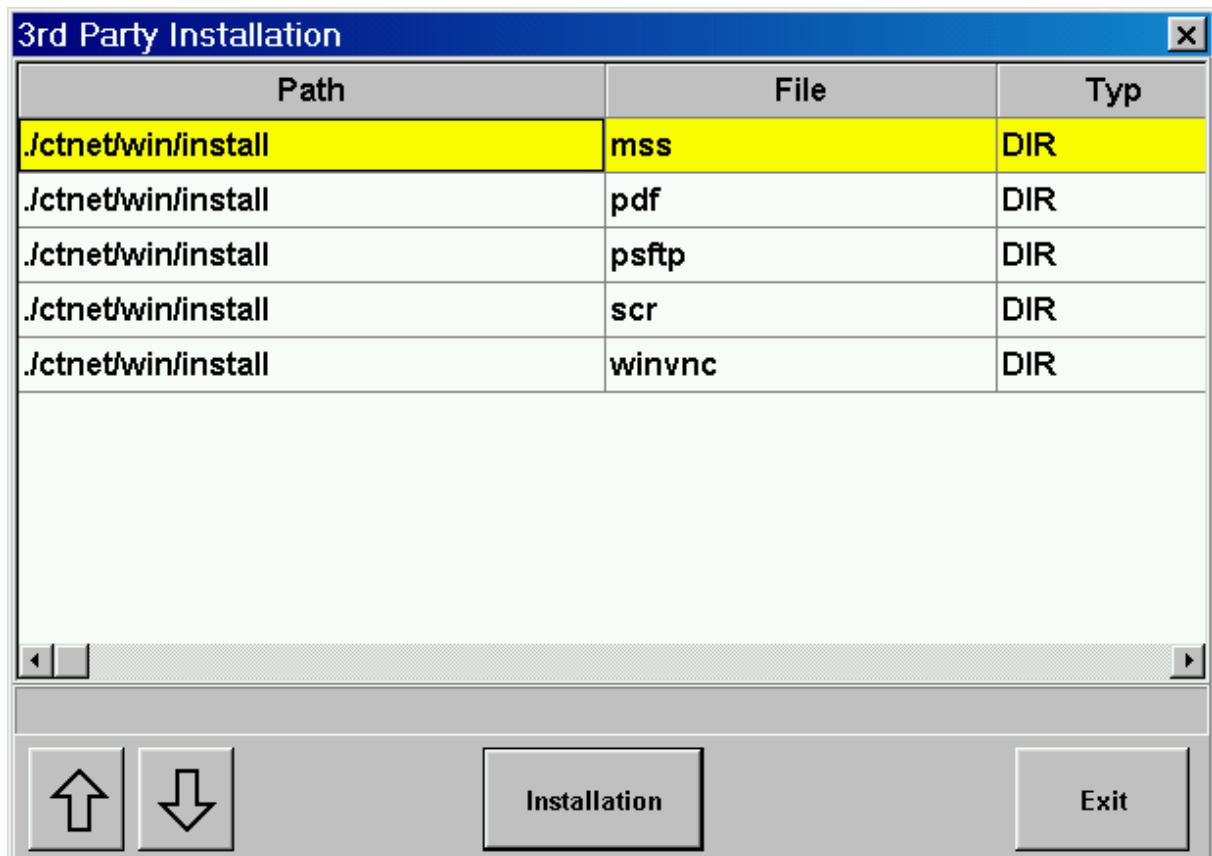
"Restore" attempts to load a backup file located in the server and then automatically restores all of the backed up files and any backed up registry entries.

As already described in the backup section, a backup file is filed in the server directory:

<hydradir>\custom\backup\ctaip\ctaipbackup2xxx.zip

Menu item "[3] Installation 3rd Party"

Please note: Possible with file server hyfserv 7.2.1.29 only



"Installation" button:

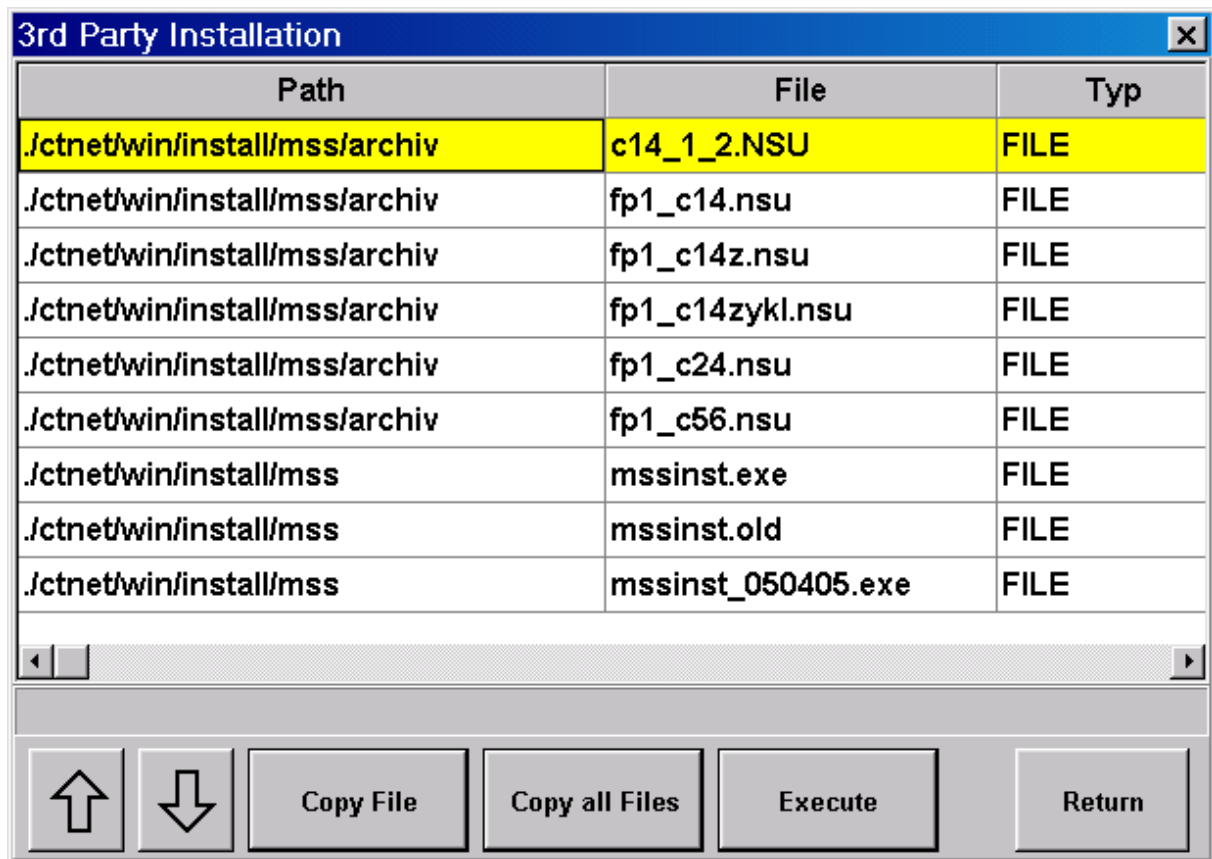
A list with all directories starting with the **<hydradir>\ctnet\win\install** directory is displayed. The directories found will be offered for selection in a dialog. By confirming a directory, this directory is downloaded and its contents are shown.

The **<hydradir>\ctnet\win\install** directory can therefore be expanded to include additional directories.

Content of a directory selected beforehand

3rd Party Installation		
Path	File	Typ
.\ctnet\win\install\mss\archiv	c14_1_2.NSU	FILE
.\ctnet\win\install\mss\archiv	fp1_c14.nsu	FILE
.\ctnet\win\install\mss\archiv	fp1_c14z.nsu	FILE
.\ctnet\win\install\mss\archiv	fp1_c14zykl.nsu	FILE
.\ctnet\win\install\mss\archiv	fp1_c24.nsu	FILE
.\ctnet\win\install\mss\archiv	fp1_c56.nsu	FILE
.\ctnet\win\install\mss	mssinst.exe	FILE
.\ctnet\win\install\mss	mssinst.old	FILE
.\ctnet\win\install\mss	mssinst_050405.exe	FILE

Having clicked one of the buttons **"copy file"** or **"copy all files"**, a selection screen opens where a directory may be chosen. The selected file or all files displayed are copied into this directory. In order to copy a single file, first it needs to be selected in the list.



Execute button: A file from the list may be executed by clicking this button. The execution program defined in Windows is used to display or execute the selected file.

12.1 Using the functions in Windows 7

Availability or feasibility of individual functions of the "INST32" installation function using Windows 7 depends on the user type:

User Type	Local User Member of the "administrators" group	Local administrator	Local User, Member of the "users" group
INST32 Function			
Updating INST32 from the HYDRA server	Yes	Yes	Yes
Installation of fonts	Yes	Yes	No
Loading the application (download)	Yes	Yes	Yes
Changing configuration settings	Yes	Yes	Yes
Installation MSS Loader	Yes	Yes	No
Installation VNC	Yes	Yes	No
Backup	Yes	Yes	Yes
Restore	Yes	Yes	Yes
Installation 3rd party components	Yes	Yes	No

User Type	Local User Member of the "administrators" group	Local administrator	Local User, Member of the "users" group
INST32 Function			
Deleting files	Yes	Yes	Yes
Starting MSS Loader	Yes	Yes	No
Starting hardware test	Yes	Yes	No
Starting reader test	Yes	Yes	No
Starting LAN test	Yes	Yes	Yes

12.2 Installation of font types in Windows 7

If the "Windows 7" operating system is used the required font types can no longer be installed by the application itself.

The required fonts have to be installed once using the above-mentioned installation program "INST32", menu item "HYDRA Fonts install". This is required to make sure the information is properly displayed on the terminal.

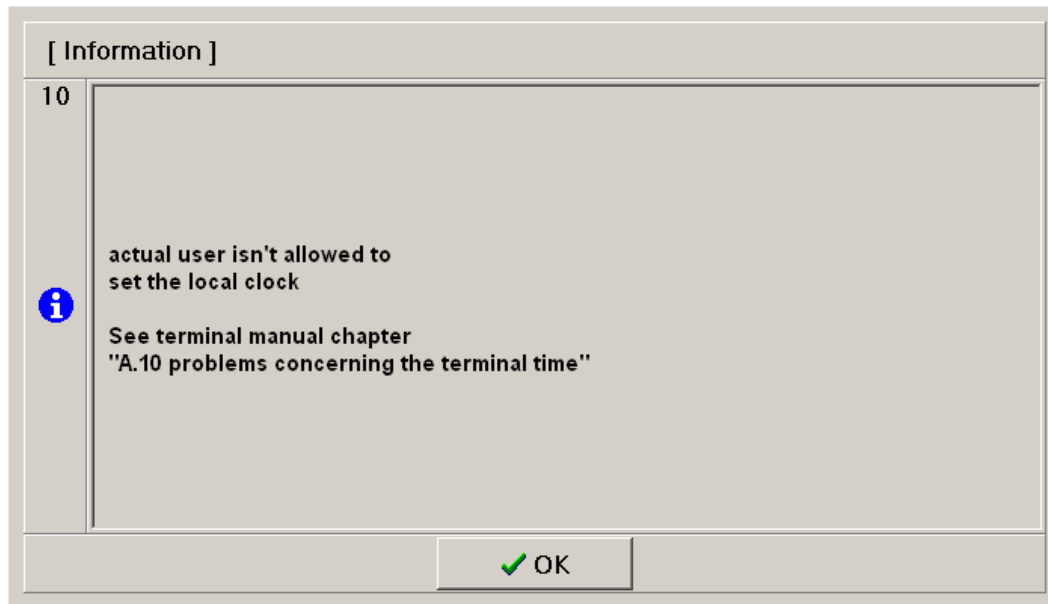
The following minimum versions are required if AIP is used with Windows 7.

Program	Version/date
inst32.exe	7.0.0.7
ctaip.exe	2.0.2.10
hydra-fonts.msi	2010-05-31

12.3 Date/time synchronization at the terminal

The HYDRA terminal software automatically synchronizes the time of the local terminal PC with the time of the HYDRA server. Usually, different Windows versions require administrative rights to be able to set the time locally.

The message below is displayed once, every time the program is started (as of version V# 2.0.2.10) if the respective Windows user does not have the required rights.



The following entry in the [system] section of the local configuration file CTAIP.INI prevents the time from being synchronized with the HYDRA server and, as a result, the above-mentioned error message is also suppressed.

```
[system]
NoLocalWatchUpdate=on
NoLocalWatchWarning=on
```

But if this configuration is enabled, the user has to make sure that the time of the HYDRA terminal PC is synchronized with another reliable source and that the time of the HYDRA terminal PC and the time of the HYDRA server are synchronous.

The user is responsible for disabling the time synchronization function. Wrong time stamps at the terminal may lead to unintentional problems when processing data, e.g. incomplete or wrong reports and evaluations or errors in posting times and durations.

12.4 Control of special watchdog hardware

New watchdog types are connected by a driver DLL, which is configured in ctaip.ini.

```
[DLL]
WdDLL=<Driver DLL>
```

Watchdog	Configuration	Required files
AEC6810	WdDLL= aaeondrv.dll	aaeondrv.dll aaeonWdt.dll

Watchdog	Configuration	Required files
		aaeonwrapper.dll

A watchdog may also be activated within the registry. The following entry has to be set:

HKEY_LOCAL_MACHINE\SOFTWARE\MPDV\CT\WdDLL=<Driver DLL>

In case both entries are set, the entry in the ctaip.ini file takes priority.